

# Metal Forming

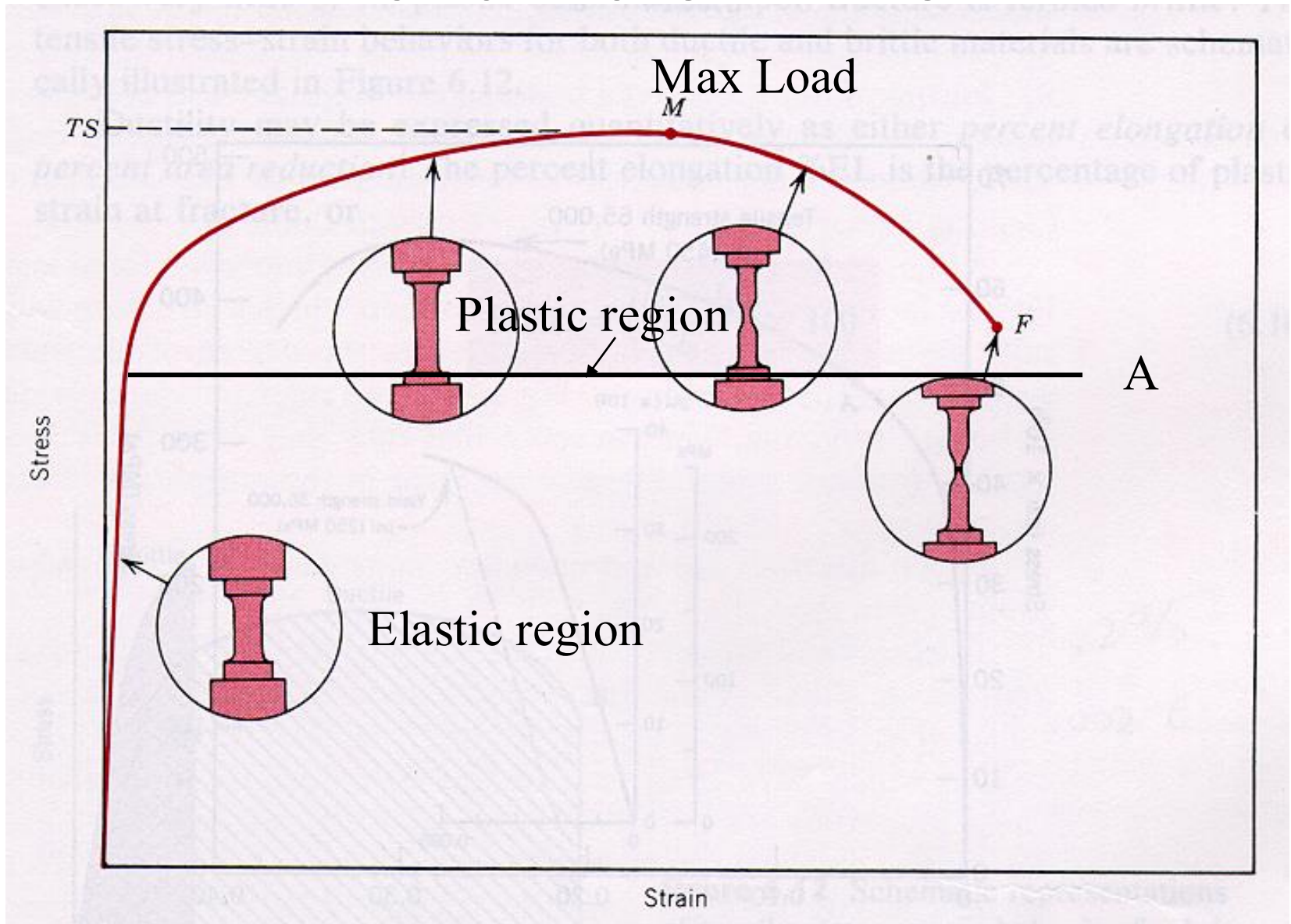


# Outline

- Fundamentals of Metal Forming
- Bulk deformation
- Sheet metalworking



# Plastic Deformation



# Forming Operations

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## Hot Working

- Above recrystallization temperature
- Allows for large deformations
- Less energy required than for cold working
- May result in surface oxidation

## Cold Working

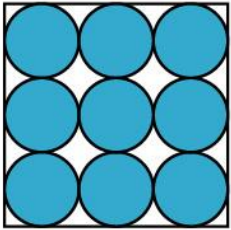
- Below recrystallization temperature
- Increases strength, but decreases ductility and corrosion resistance
- Better control of dimensions
- Better surface finish

## Temperature in metal forming

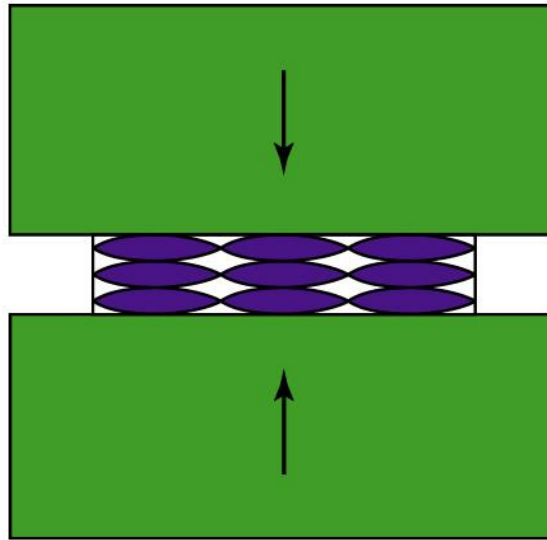
- Better accuracy
- Better Surface finishing
- Strain hardening increase strength
- Costs

# Preferred Orientation

(a)

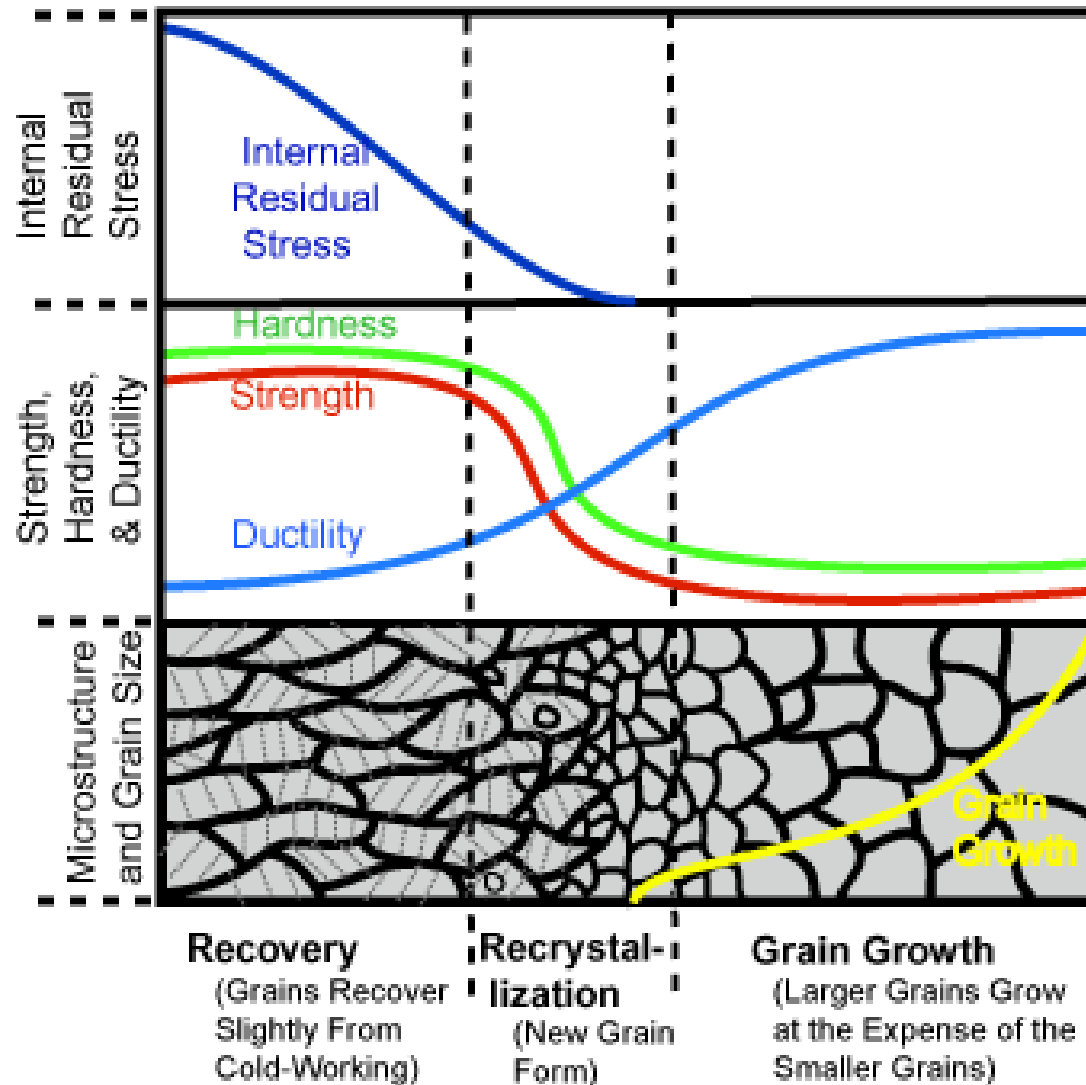


(b)



Plastic deformation of idealized (equiaxed) grains in a specimen subjected to compression (such as occurs in the rolling or forging of metals): (a) before deformation; and (b) after deformation. Note the alignment of grain boundaries along a horizontal direction; this effect is known as preferred orientation.

# Recrystallization



# Cold Working

- Metal forming performed at room temp or slightly above (below their recrystallisation temperature)

## Advantage

Better accuracy

Better Surface finishing

Strain hardening increase strength

Costs

## Disadvantage

Higher forces are required

Care the surface

Ductility is limited.

# Cracks

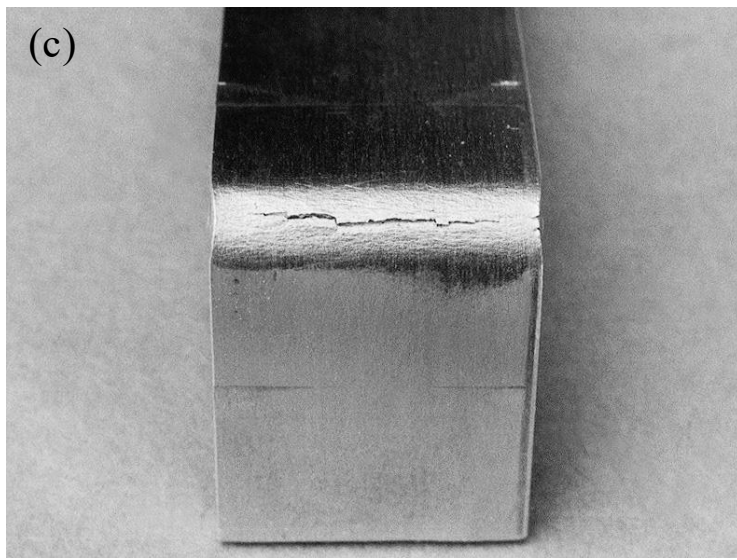
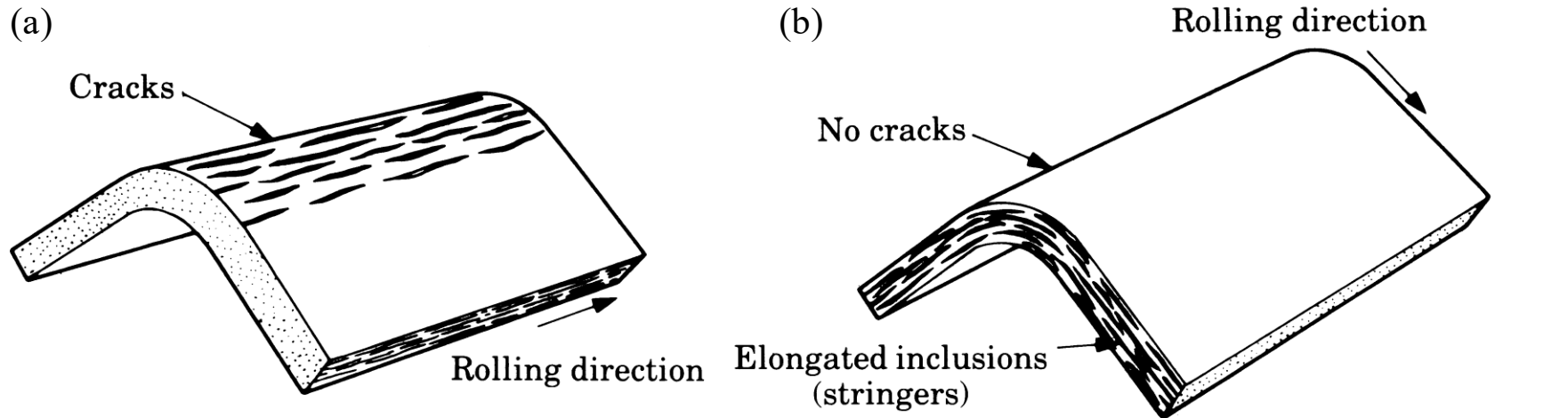


Figure (a) and (b) The effect of elongated inclusions (stringers) on cracking, as a function of the direction of bending with respect to the original rolling direction of the sheet. (c) Cracks on the outer surface of an aluminum strip bent to an angle of  $90^\circ$ . Note the narrowing of the top surface due to the Poisson effect.

# Hot working

- T above Recrystallization Temp.
- $0.5T_m < T < 0.75T_m$

## Advantage

- The shape can be significantly altered
- Lower forces are required
- Fracture can be reduced
- Strength properties are generally isotropic
- No strengthening of the part occurs from work hardening

## Disadvantage

**Lower dimensional accuracy, higher energy, poorer surface finish**  
**Shorter tool life**

# Warm Working

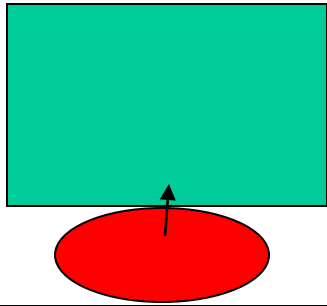
- $0.3T_m < T < 0.5T_m$

## **Advantage**

- Lower forces are required
- More intricate work geometries possible
- The need for annealing may be reduced

# Isothermal

- For high speed steel, titanium alloy, good hot hardness



Quick heat transfer

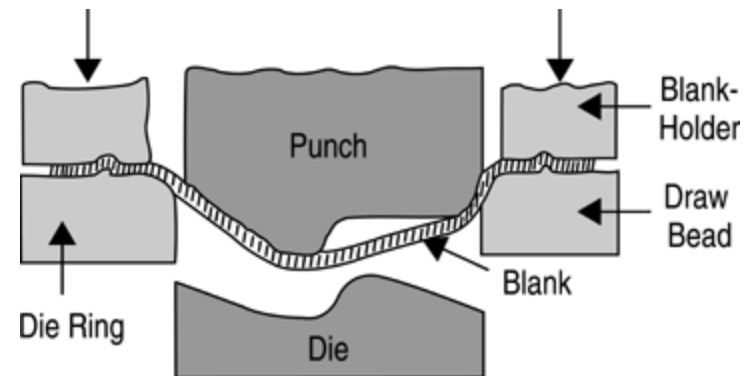
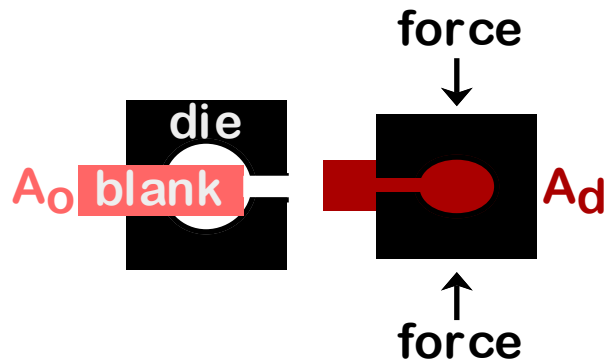
**- Raising strength leading to high residual stress**



**Cracking**

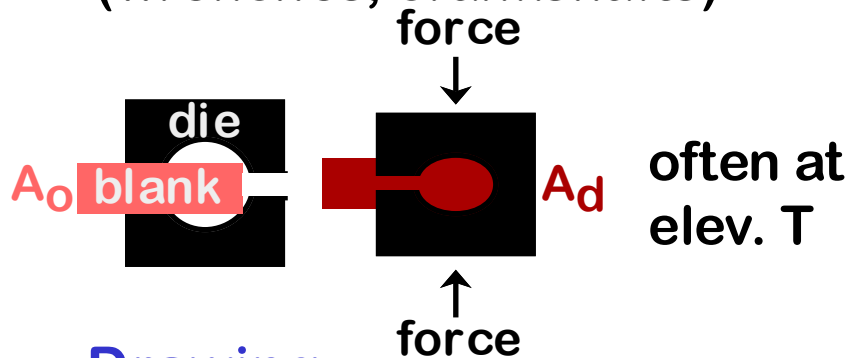
# Metal forming process

- 1. Bulk deformation processes
- 2. Sheet metal working processes

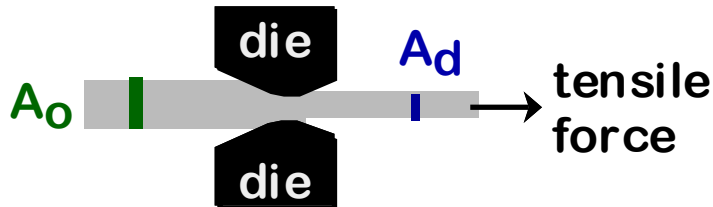


# Bulk Deforming Processes

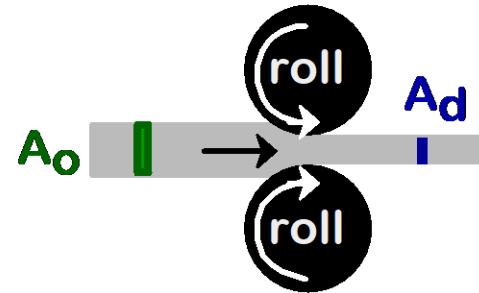
- **Forging**  
(wrenches, crankshafts)



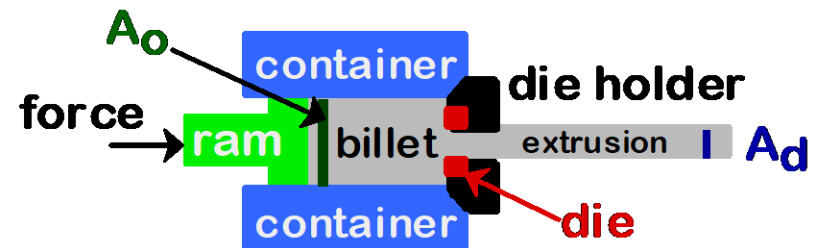
- **Drawing**  
(rods, wire, tubing)



- **Rolling**  
(I-beams, rails)

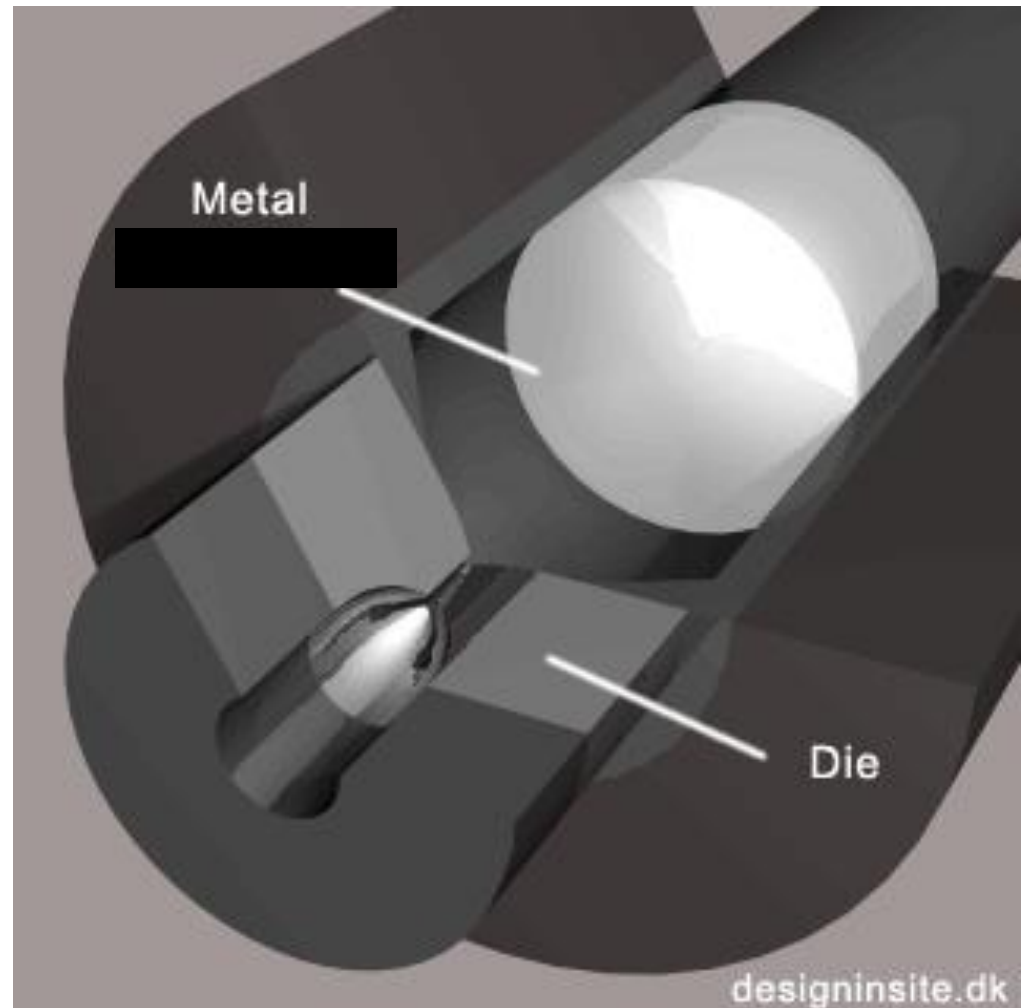


- **Extrusion**  
(rods, tubing)



# Extrusion

Metal forced/squeezed out through a hole (die)



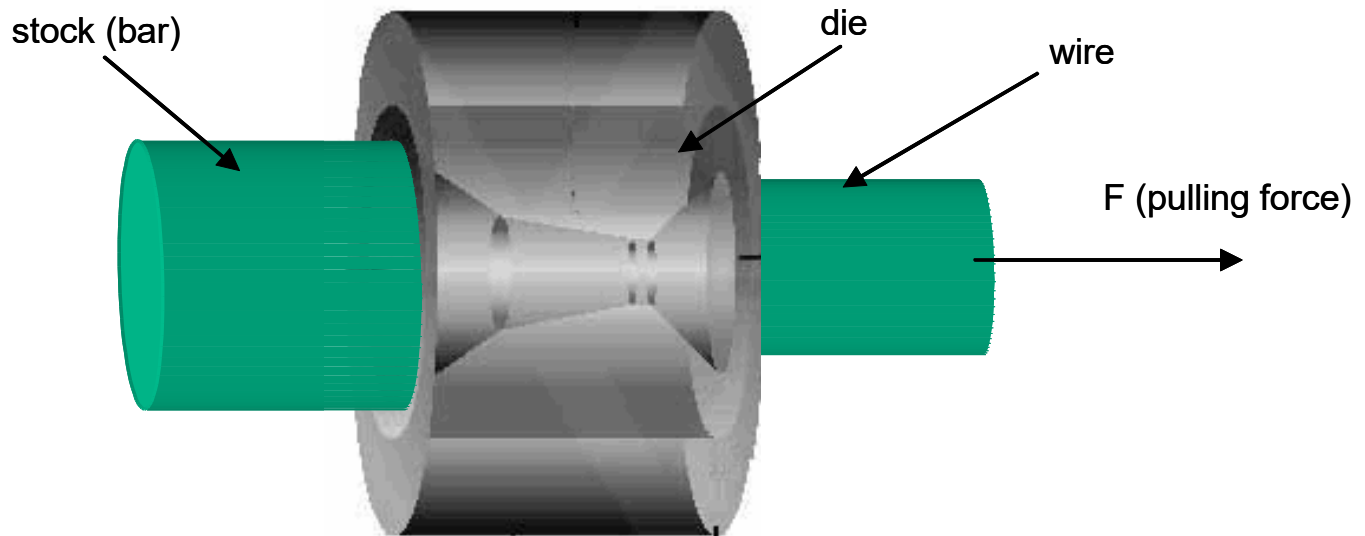
Typical use: ductile metals (Cu, Steel, Al, Mg), Plastics, Rubbers

Common products:

Al frames of white-boards, doors, windows, ...

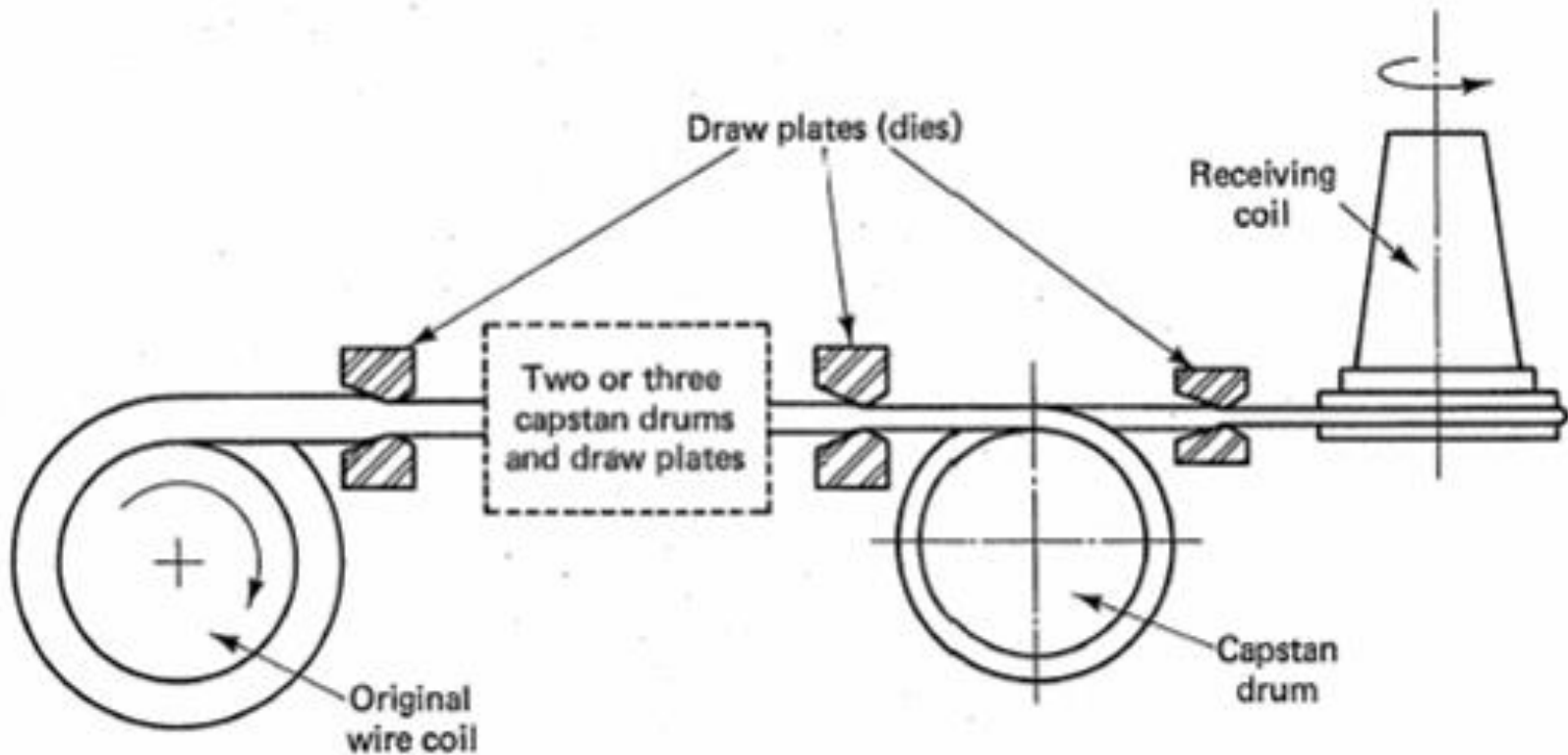
# Drawing

Similar to extrusion, except: *pulling force* is applied



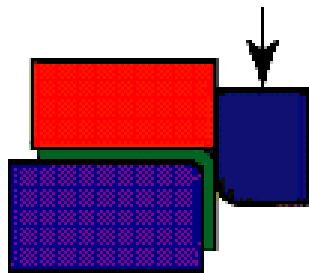
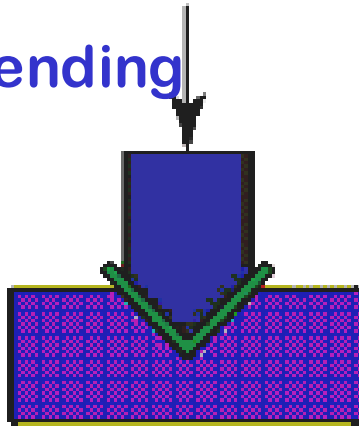
Commonly used to make wires from round bars

# Continuous Drawing

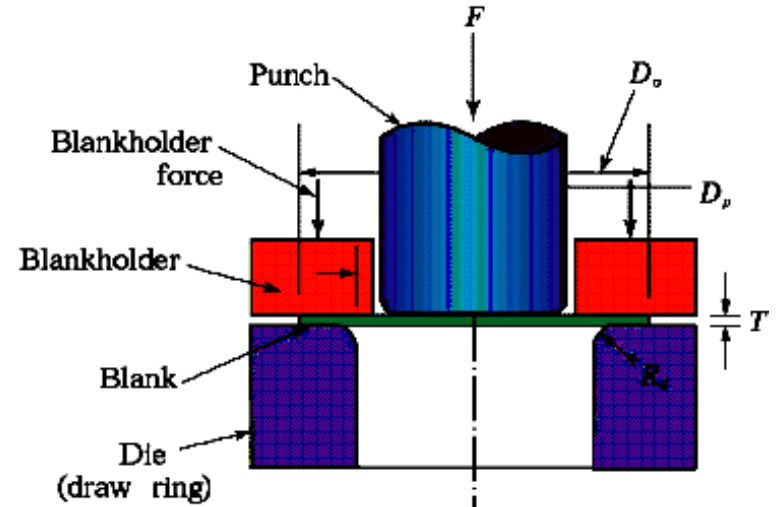


# Sheet Metalworking

- Bending



- Deep drawing



- Shearing

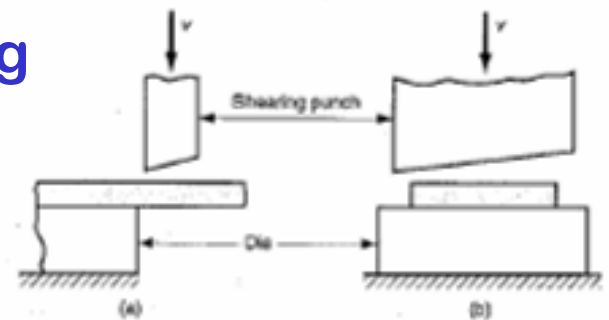
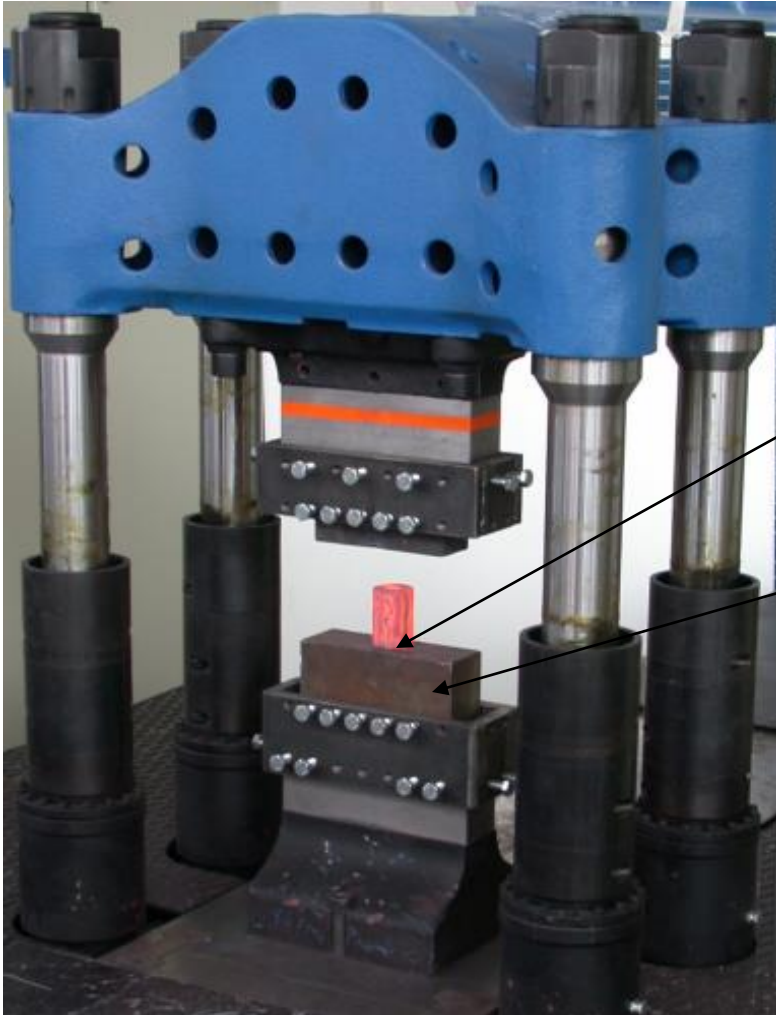


FIGURE 22.3 Shearing operation: (a) side view of the shearing operation; (b) front view of power shears equipped with inclined upper cutting blade. Symbol  $v$  indicates motion.

# Forging

[Heated] metal is beaten with a heavy hammer to give it the required shape



Hot forging,

open-die

# Forging

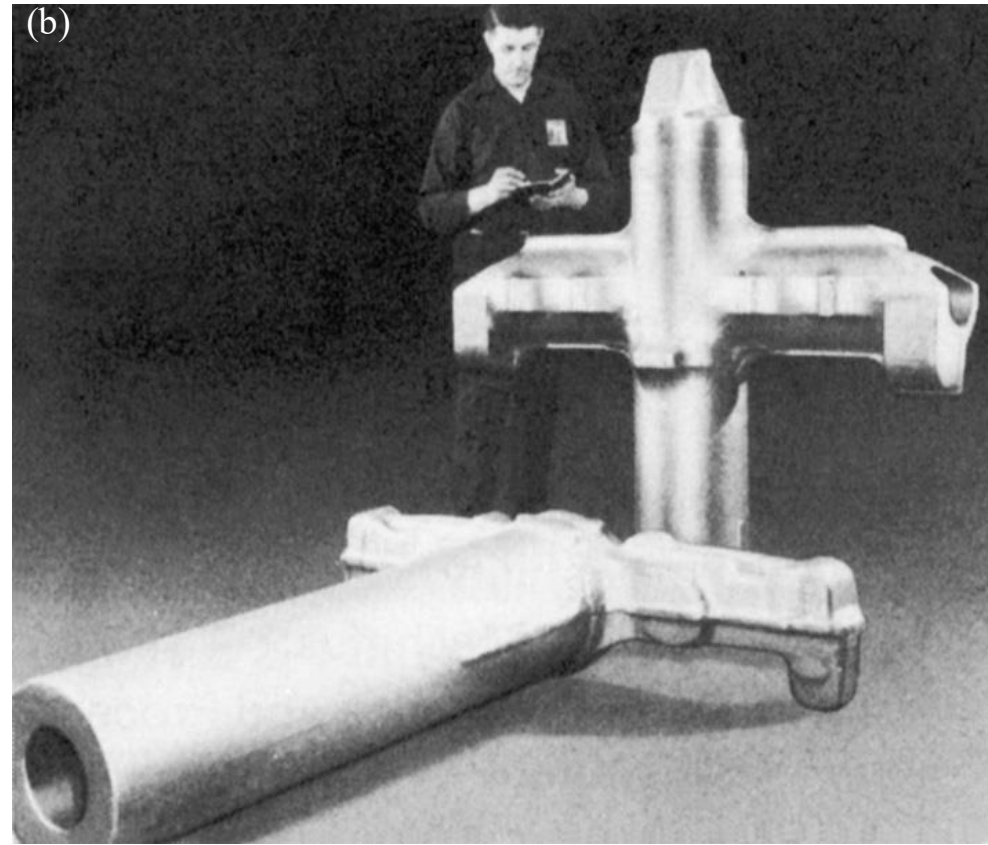
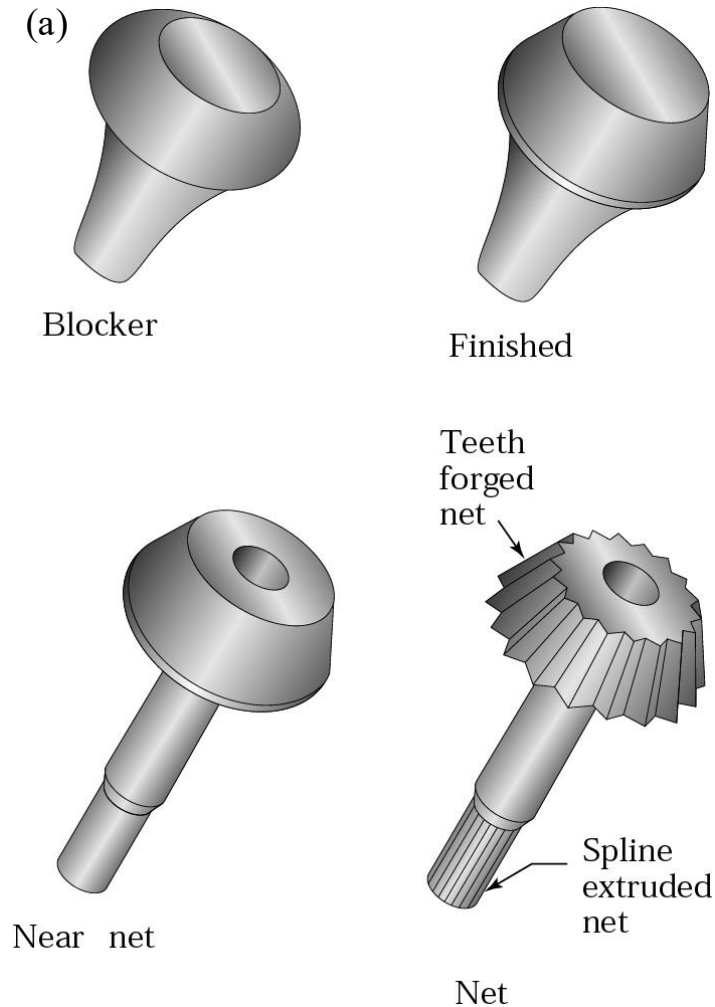
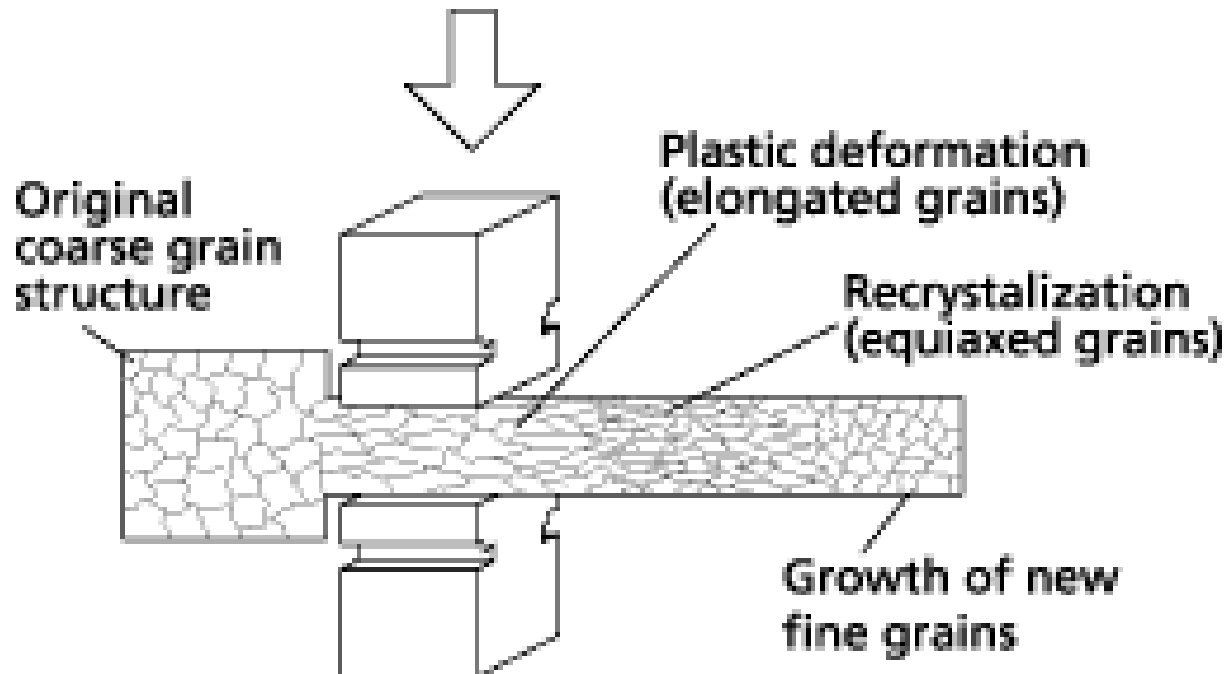


Figure 14.1 (a) Schematic illustration of the steps involved in forging a bevel gear with a shaft. *Source:* Forging Industry Association. (b) Landing-gear components for the C5A and C5B transport aircraft, made by forging. *Source:* Wyman-Gordon Company.

# Quality of forged parts

Surface finish/Dimensional control:  
Better than casting (typically)

Stronger/tougher than cast/machined parts of same material



# Grain Flow Pattern of Pierced Round Billet

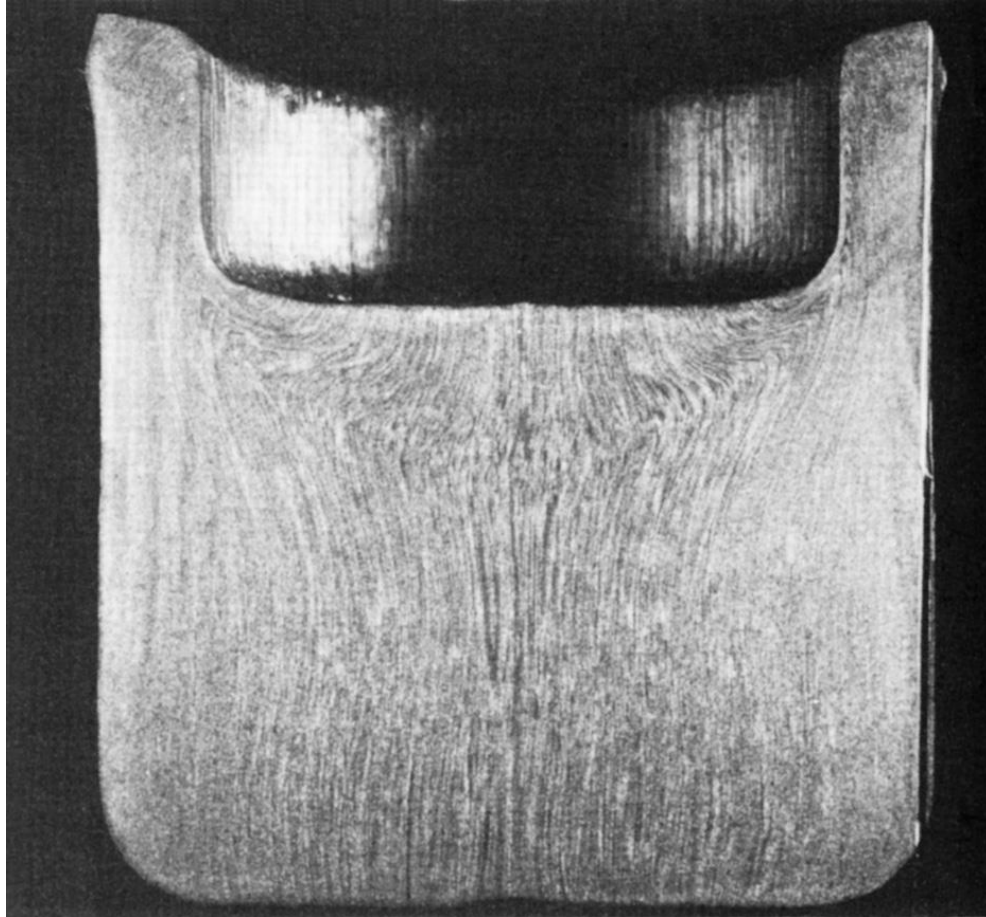


Figure 14.12 A pierced round billet, showing grain flow pattern. *Source:* Courtesy of Ladish Co., Inc.

# Grain Flow Comparison

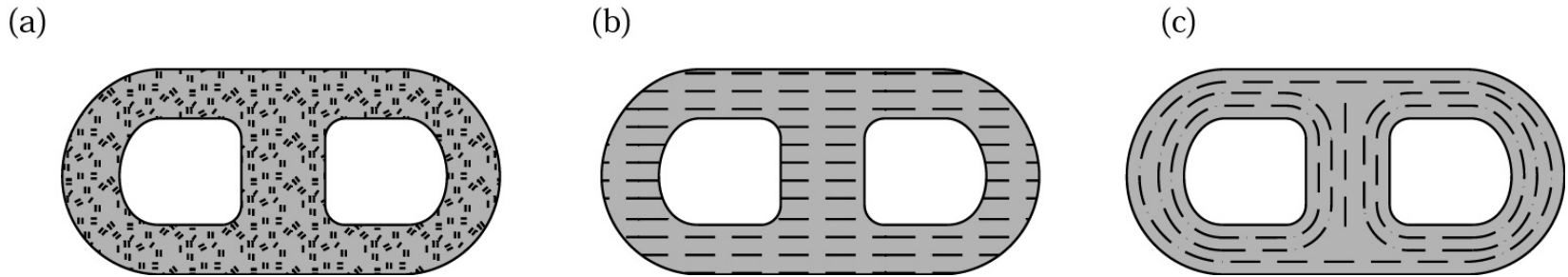


Figure 14.3 A part made by three different processes, showing grain flow. (a) casting, (b) machining, (c) forging. *Source:* Forging Industry Association.

# Upsetting

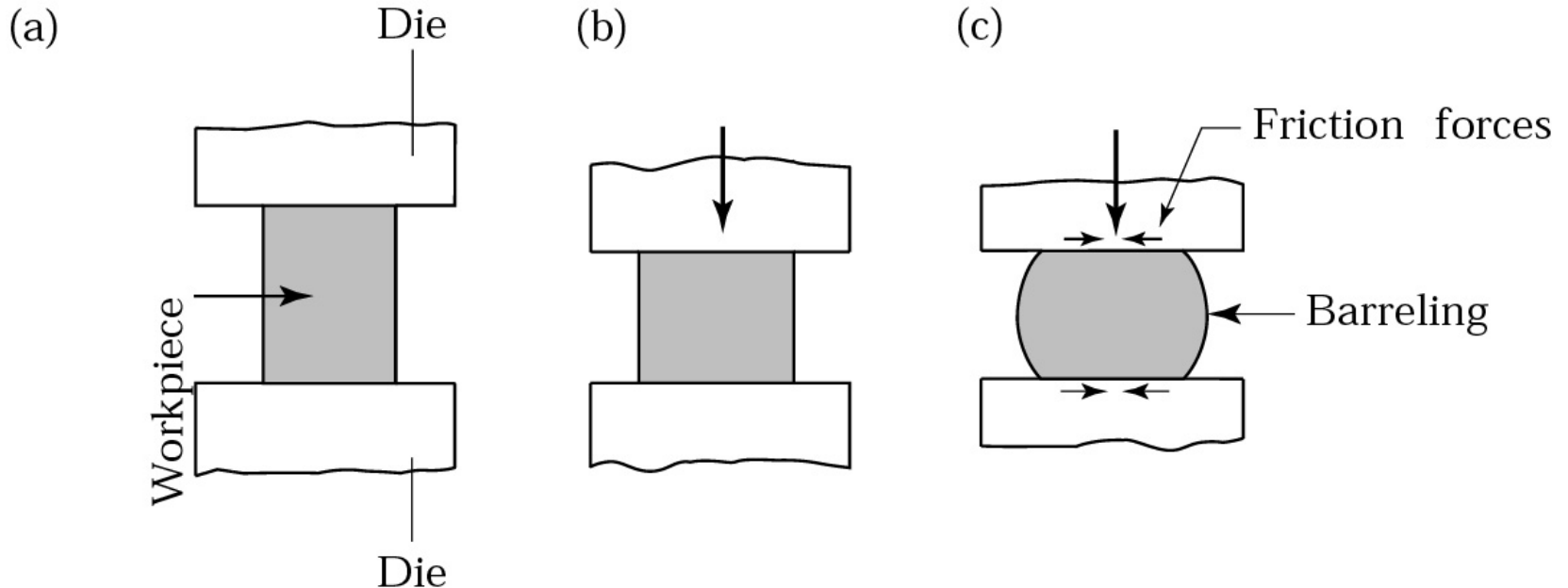


Figure (a) Solid cylindrical billet upset between two flat dies. (b) Uniform deformation of the billet without friction. (c) Deformation with friction. Note barreling of the billet caused by friction forces at the billet-die interfaces.

# Impression-Die Forging

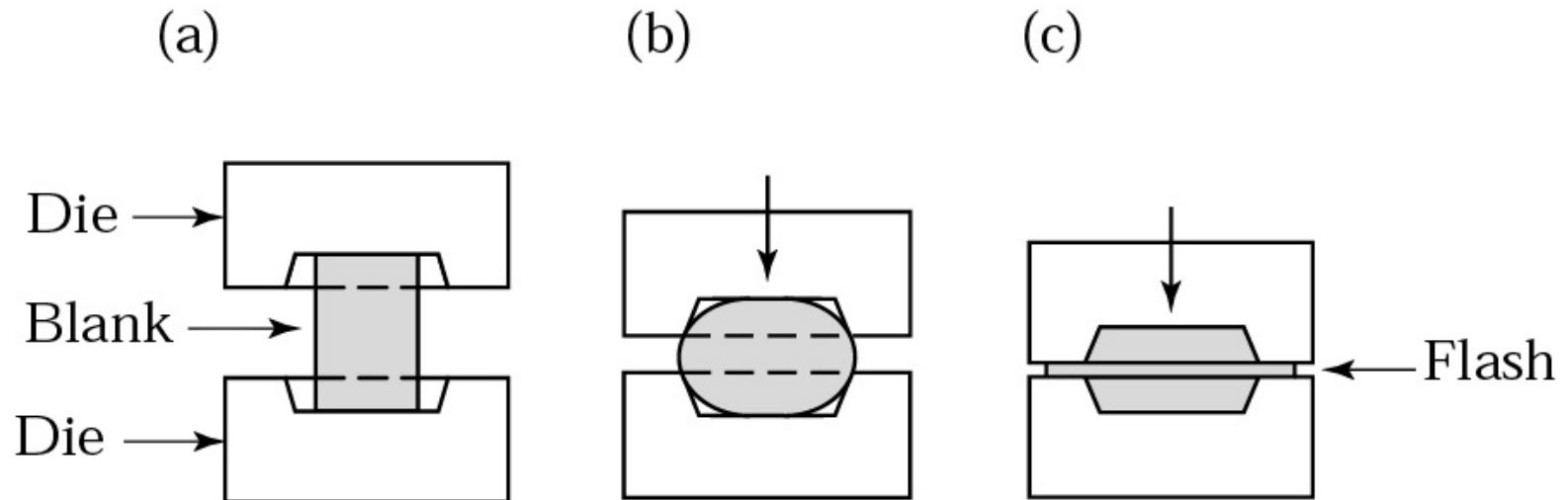
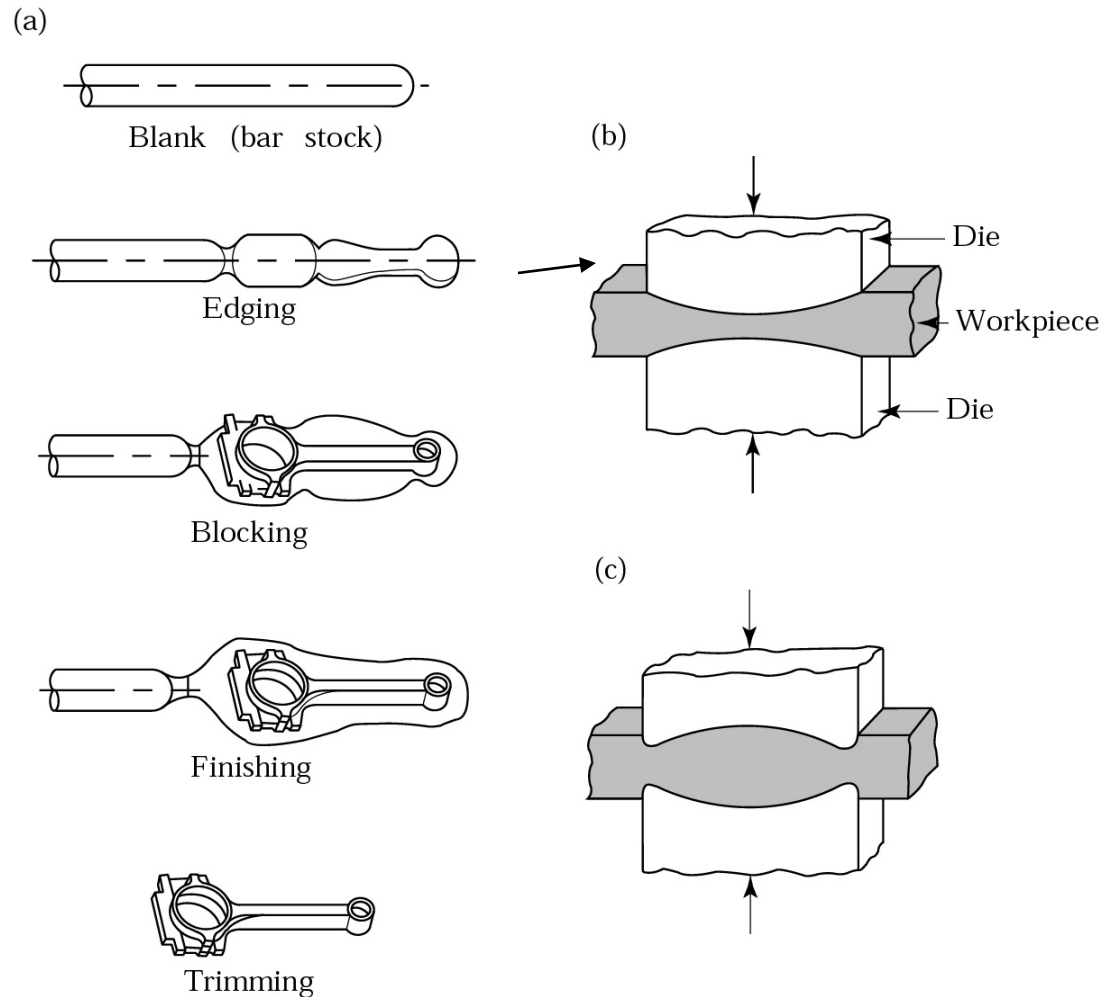


Figure 14.6 Stages in impression-die forging of a solid round billet. Note the formation of flash, which is excess metal that is subsequently trimmed off (see Fig. 14.8).

# Forging a Connecting Rod

Figure (a) Stages in forging a connecting rod for an internal combustion engine. Note the amount of flash required to ensure proper filling of the die cavities. (b) Fullering, and (c) edging operations to distribute the material when preshaping the blank for forging.



# Trimming Flash from a Forged Part

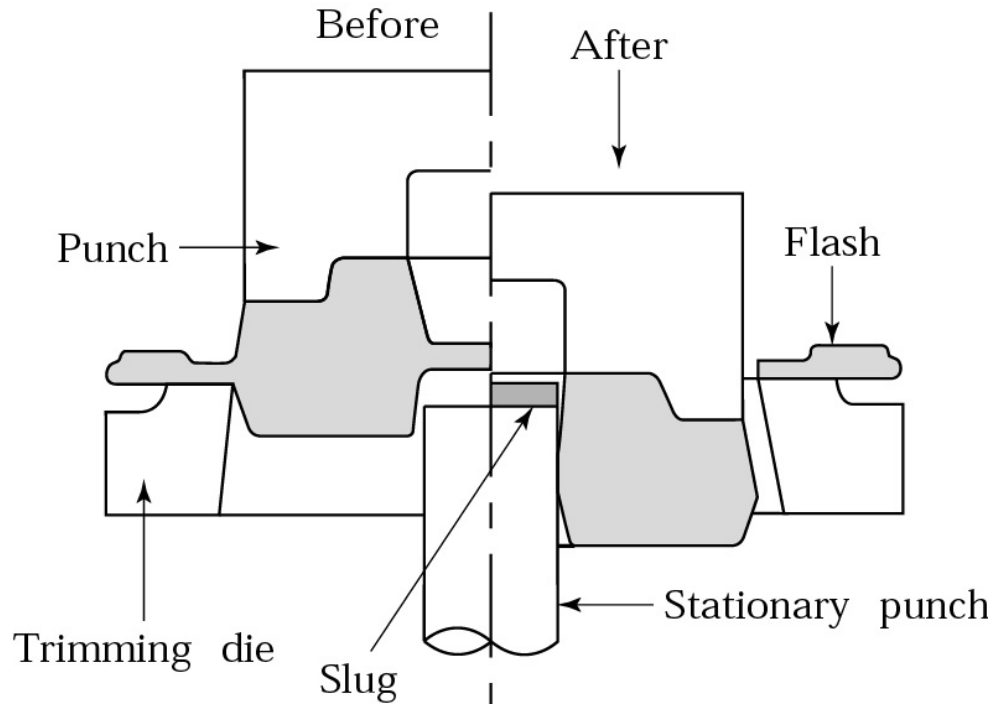


Figure. Trimming flash from a forged part. Note that the thin material at the center is removed by punching.

# Flashless forging

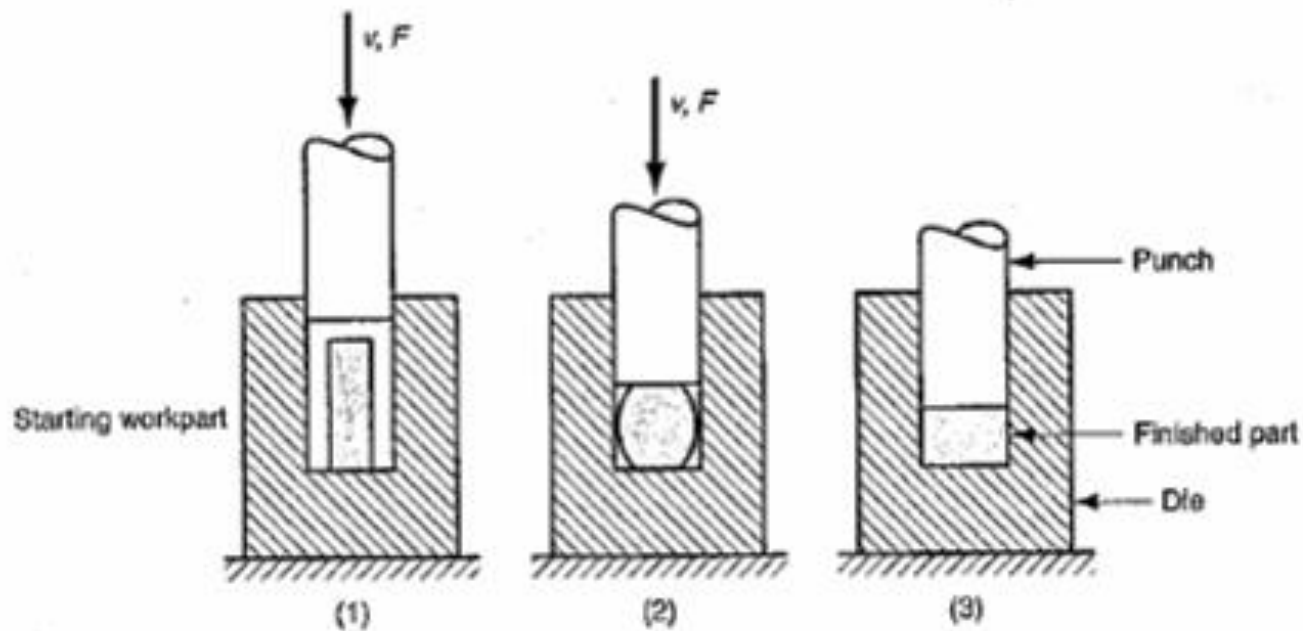
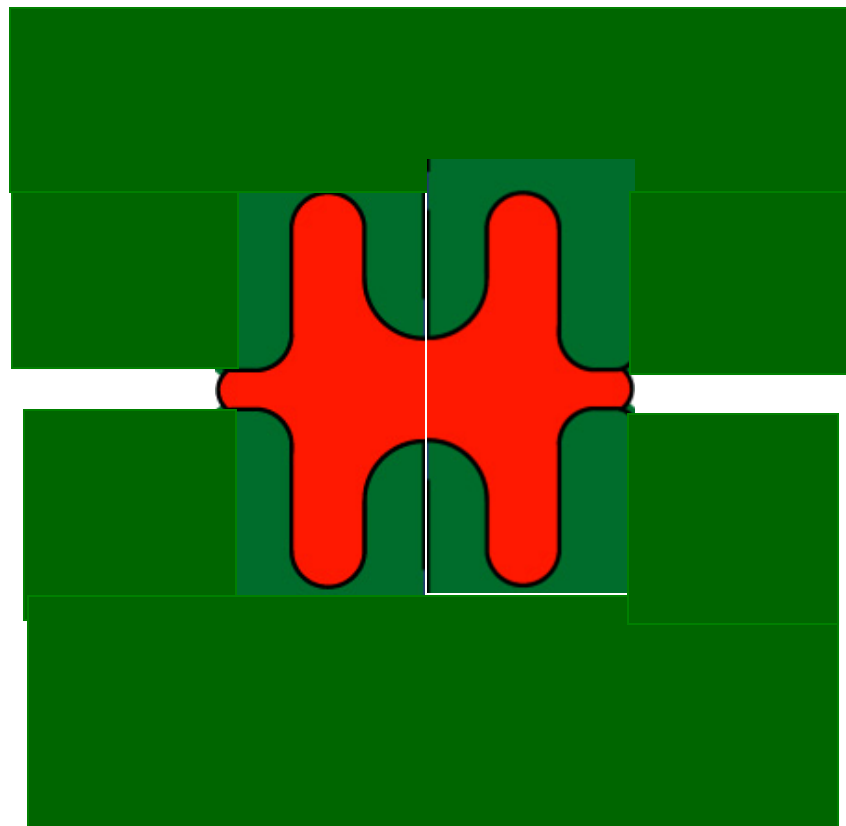
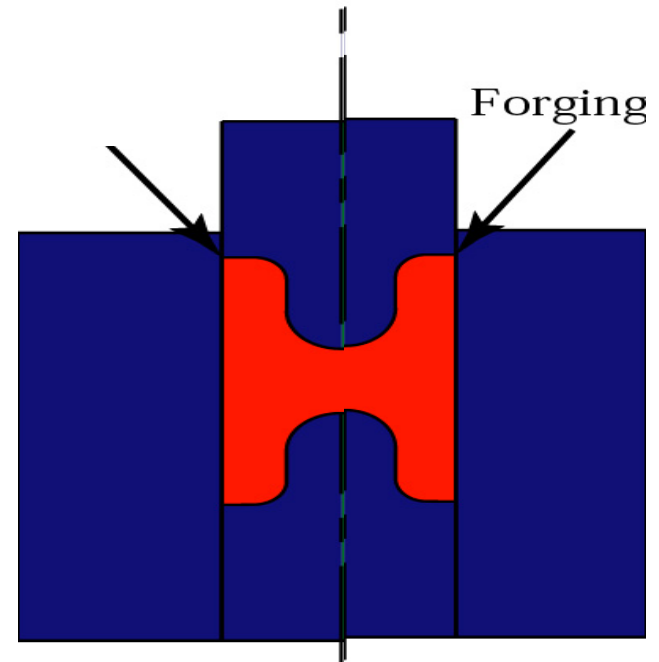
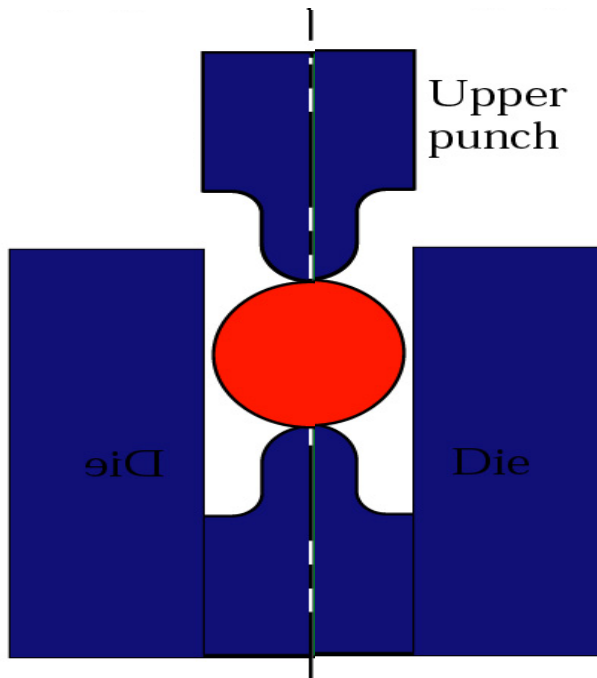


FIGURE 21.18 Flashless forging: (1) just before initial contact with workpiece, (2) partial compression, and (3) final punch and die closure. Symbols  $v$  and  $F$  indicate motion ( $v$  = velocity) and applied force, respectively.

# Normal



# Flashless Forging



of stroke

# Comparison of Forging With and Without Flash

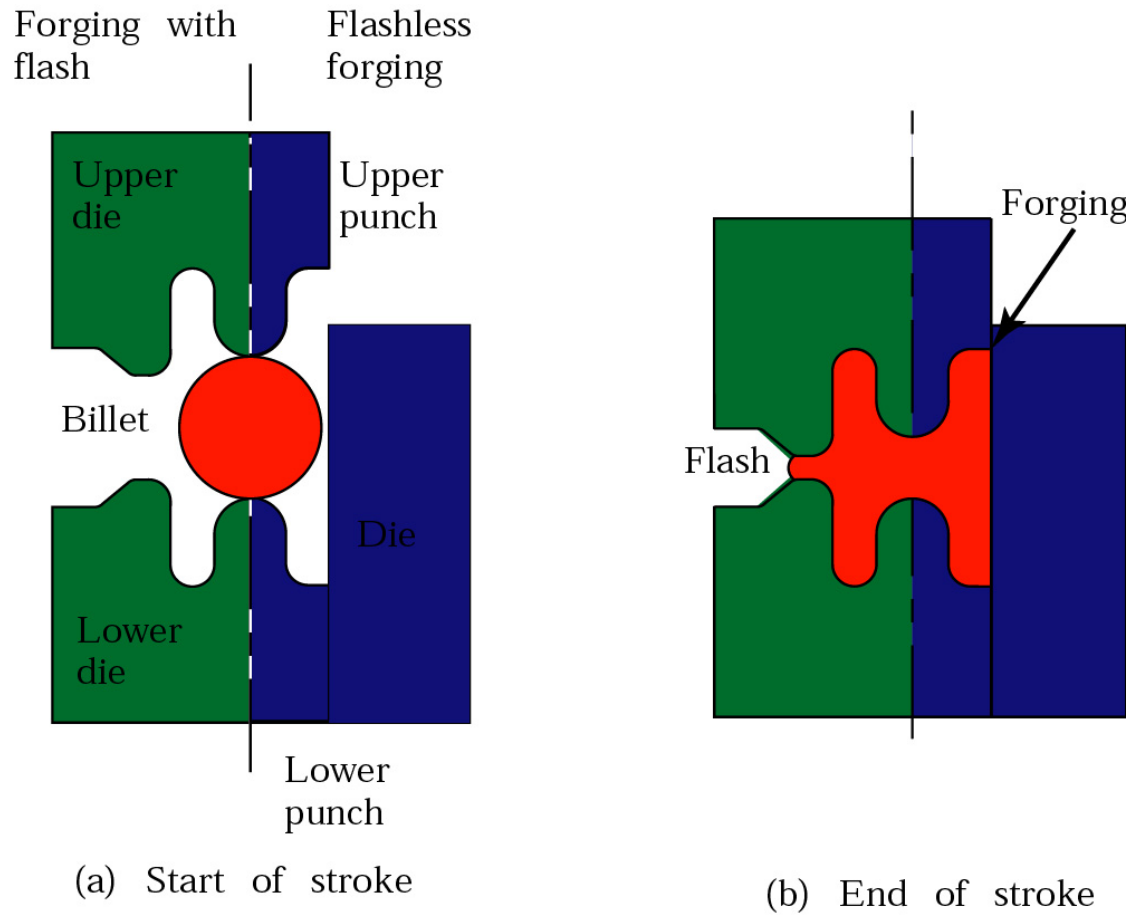


Figure 14.9  
Comparison of closed-die forging to precision or flashless forging of a cylindrical billet.  
*Source:* H. Takemasu, V. Vazquez, B. Painter, and T. Altan.

# Unit Cost in Forging

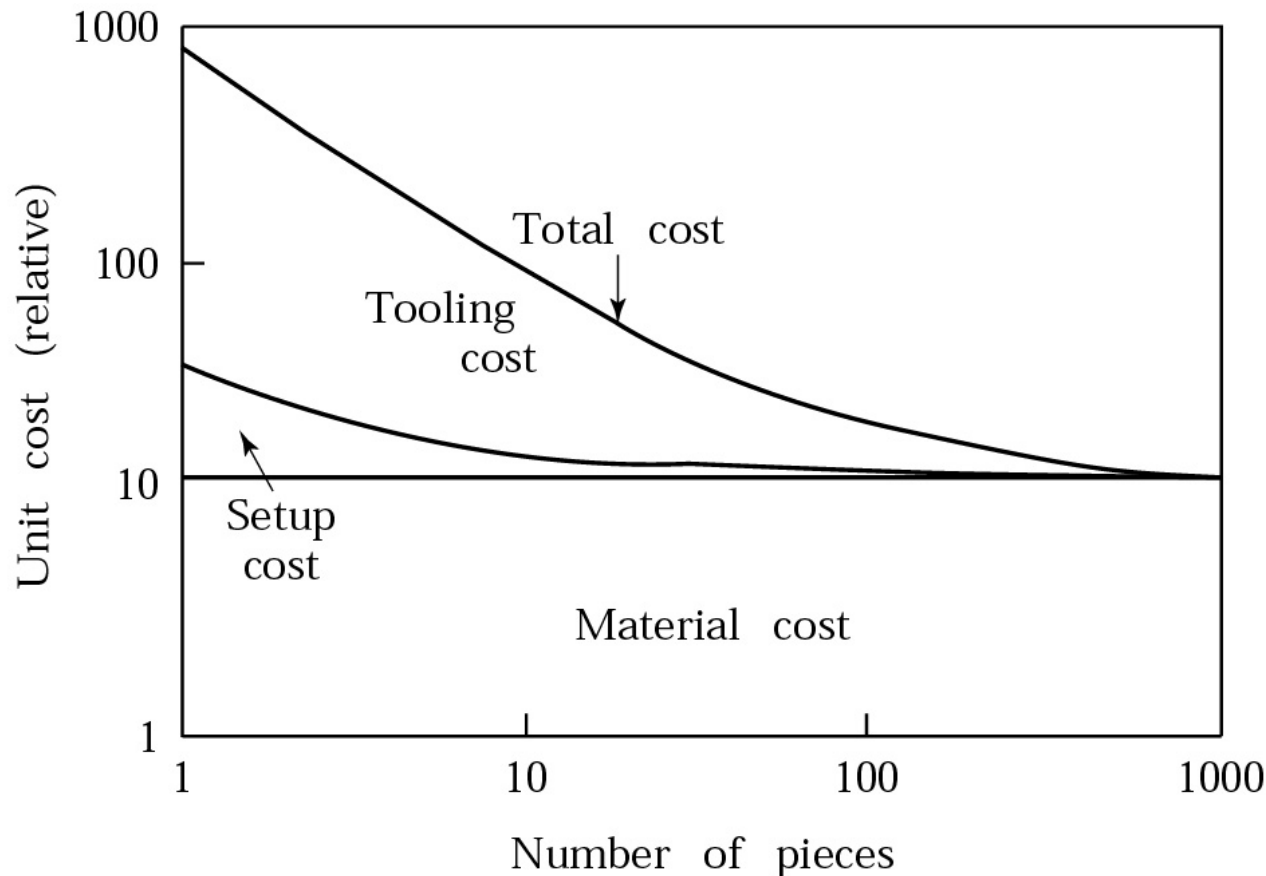
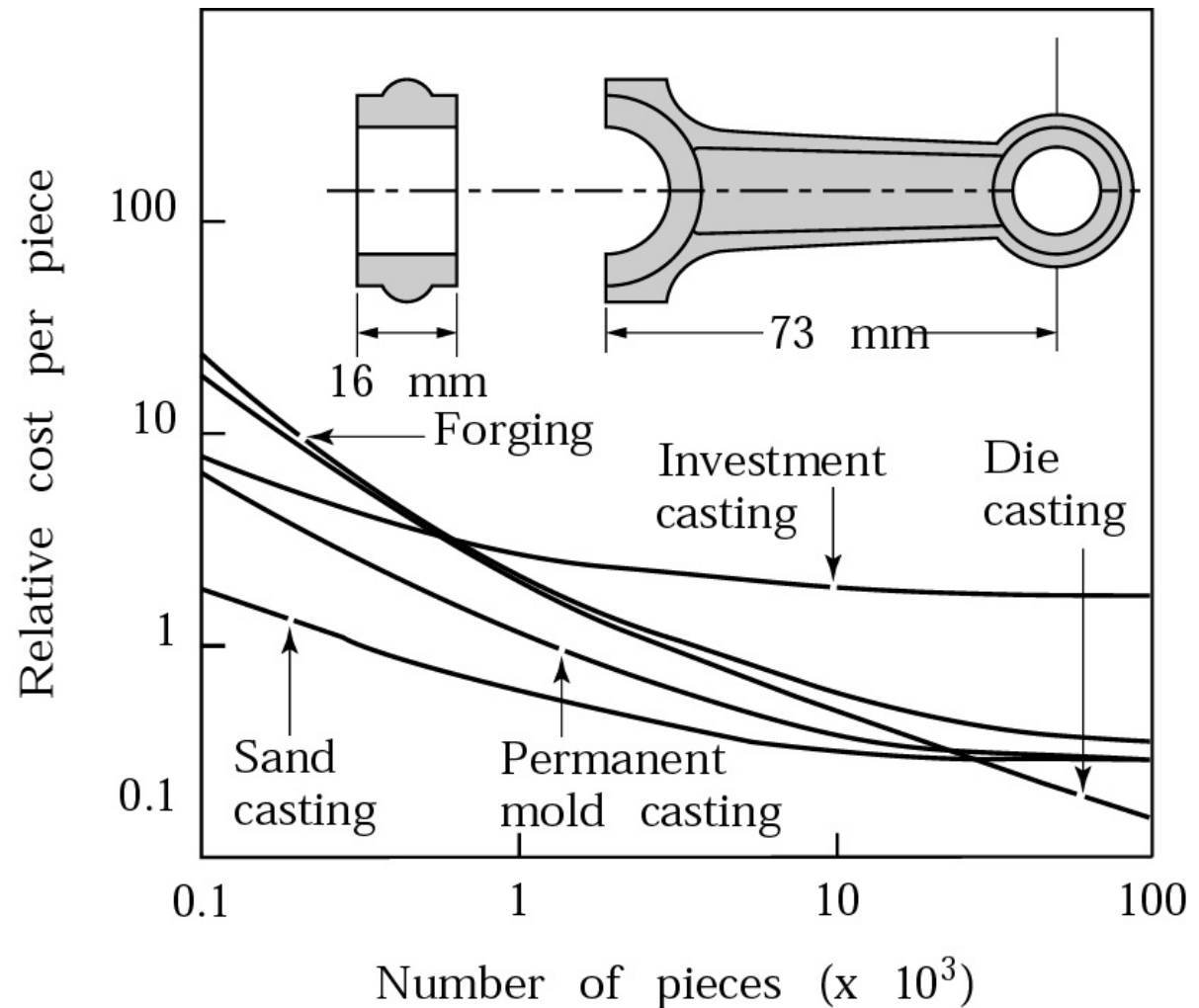


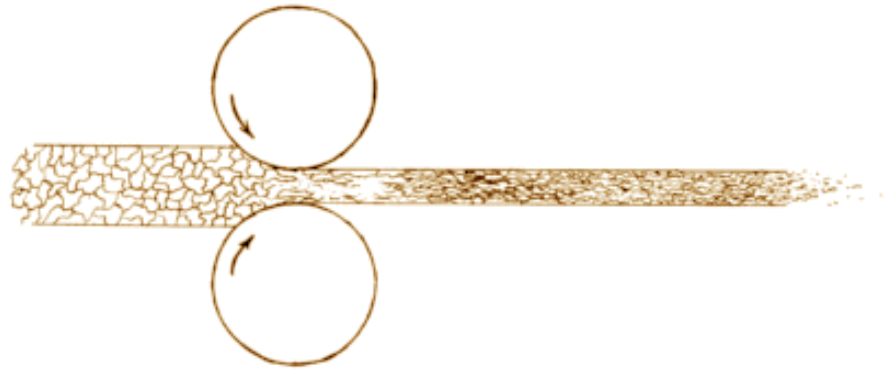
Figure Typical unit cost (cost per piece) in forging; note how the setup and the tooling costs per piece decrease as the number of pieces forged increases, if all pieces use the same die.

# Relative Unit Costs of a Small Connecting Rod

Figure .  
Relative unit costs of a small connecting rod made by various forging and casting processes. Note that, for large quantities, forging is more economical. Sand casting is the more economical process for fewer than about 20,000 pieces.



# *Rolling*

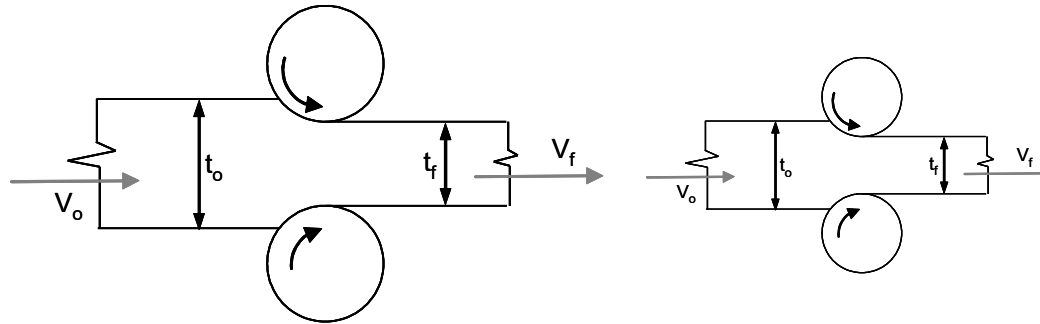


ROLLING PROCESS

## Important Applications:

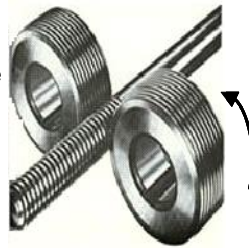
Steel Plants,  
Raw stock production (sheets, tubes, Rods, etc.)  
Screw manufacture

# Rolling Basics



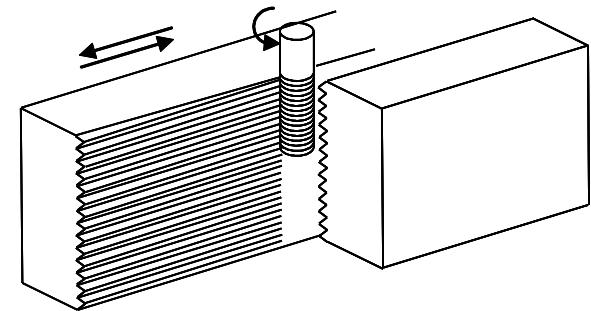
## Screw manufacture:

stationary die



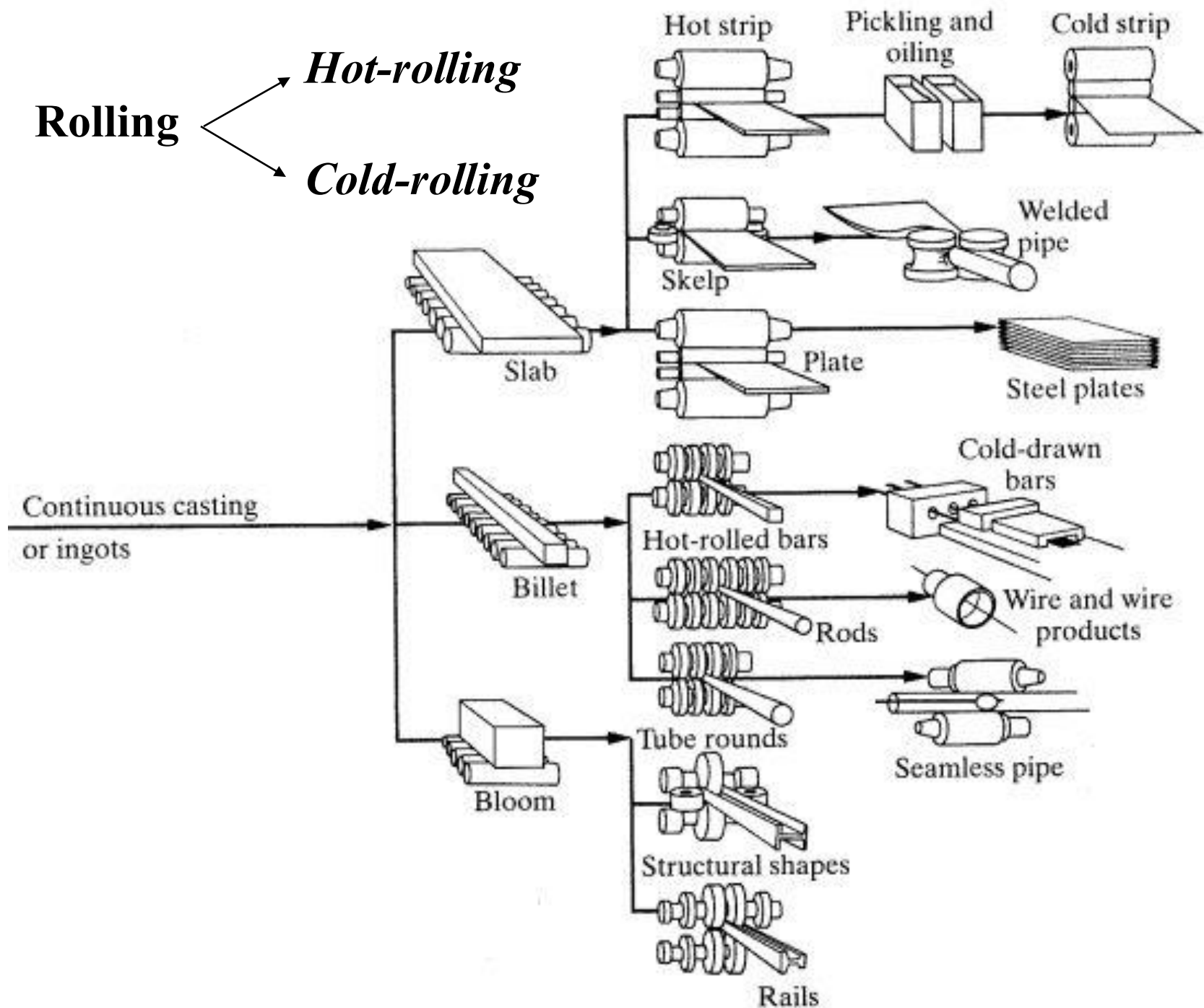
rolling die

thread rolling machine

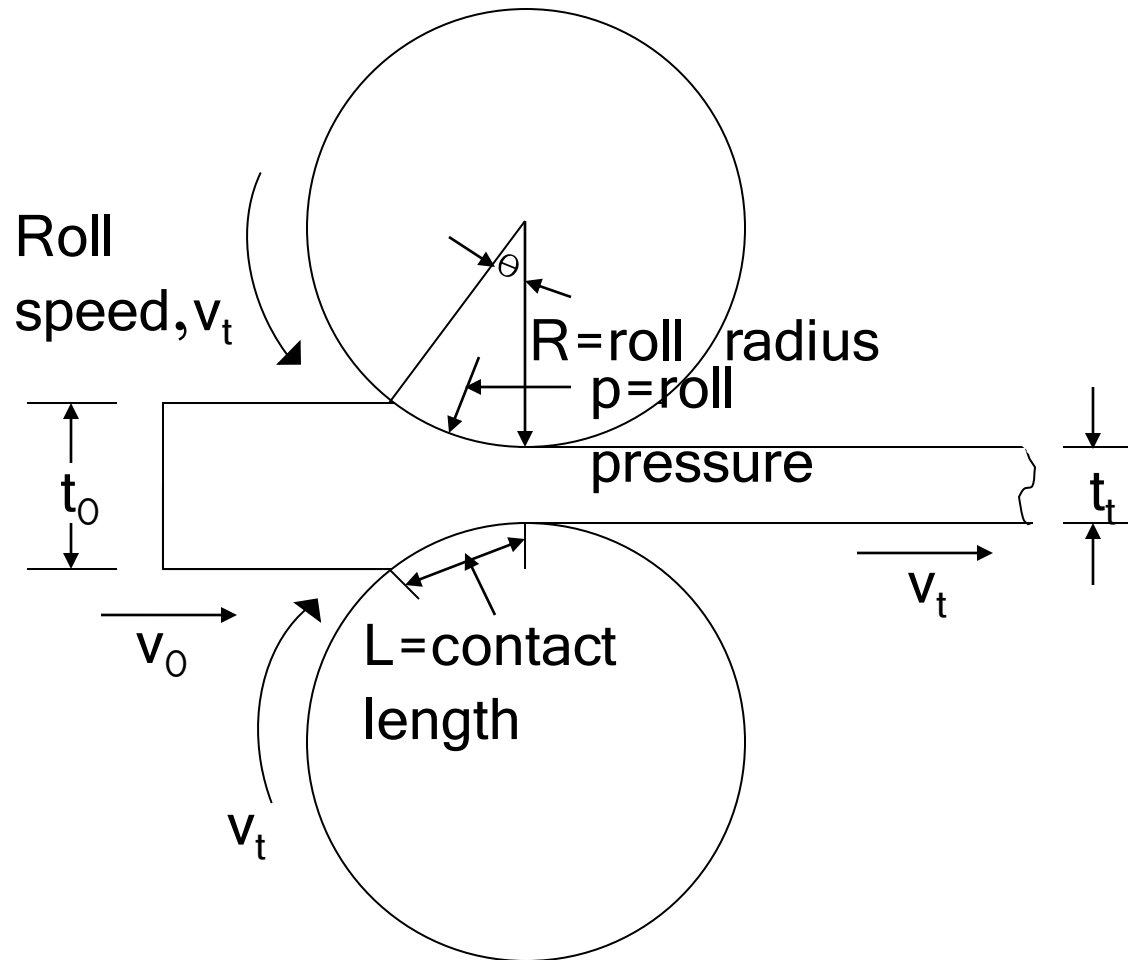


Reciprocating flat thread-rolling dies

**Rolling** *Hot-rolling*  
*Cold-rolling*



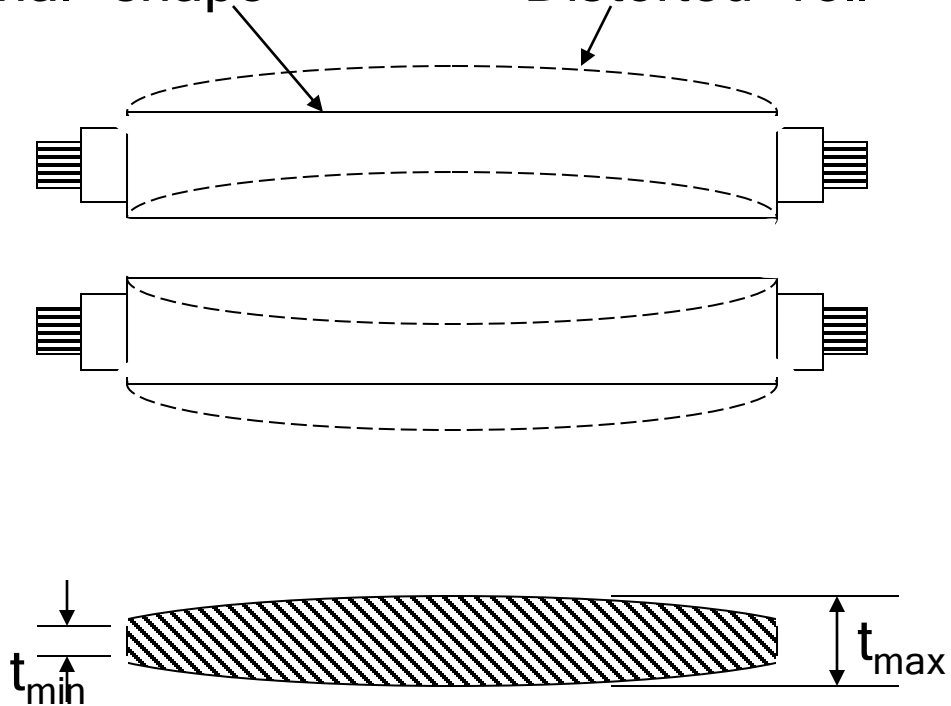
# Rolling Theory



# Distorted roll

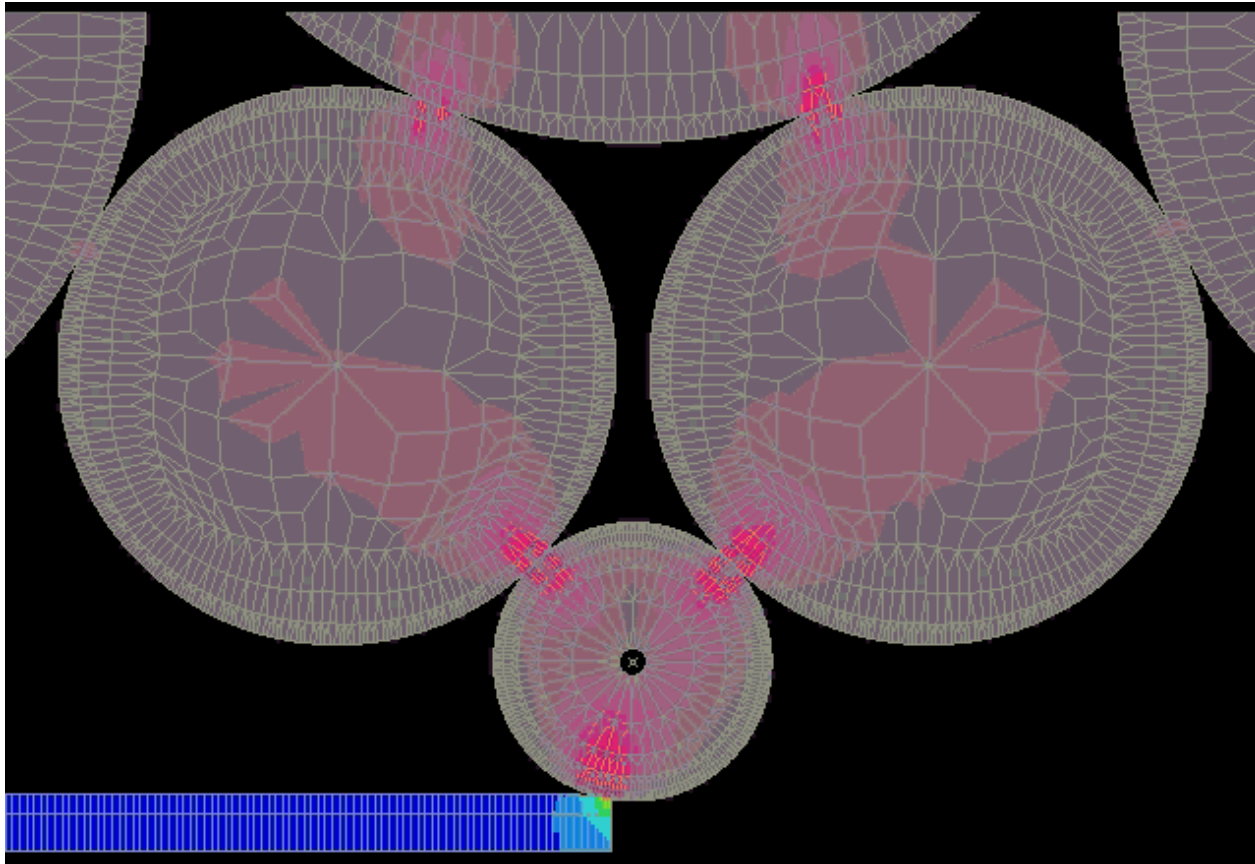
Original shape

Distorted roll

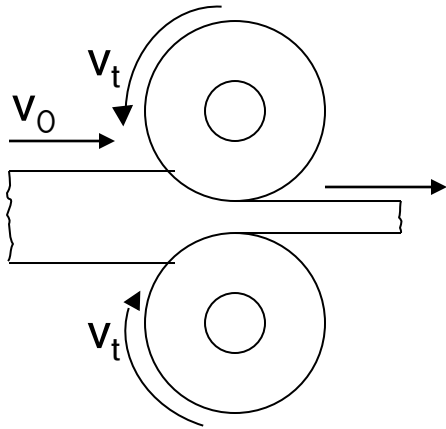


Cross section of the sheet

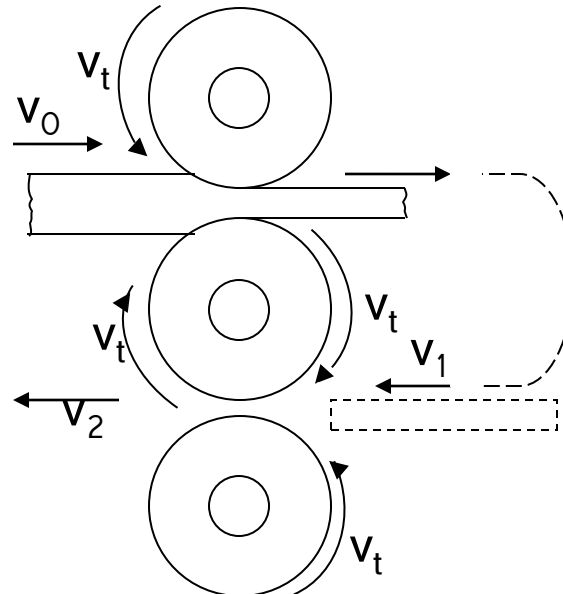
# Rolling force



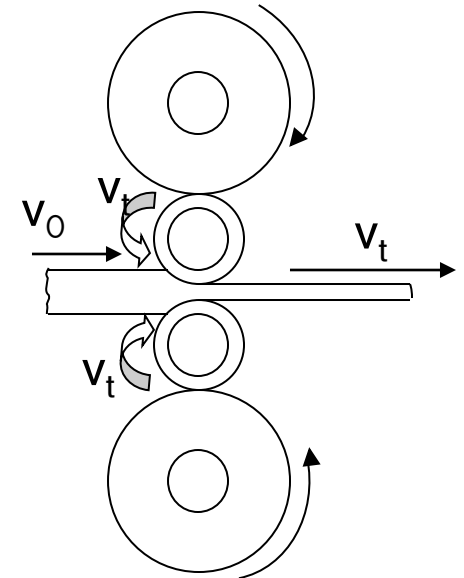
# Various Configurations



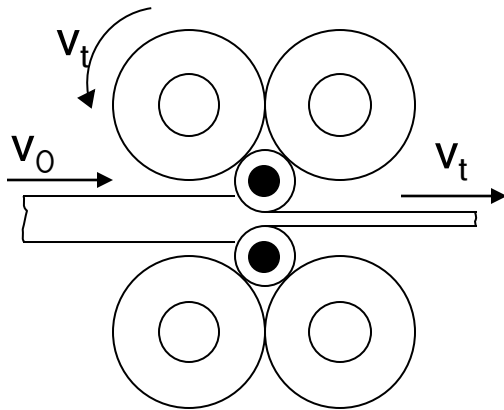
Two high



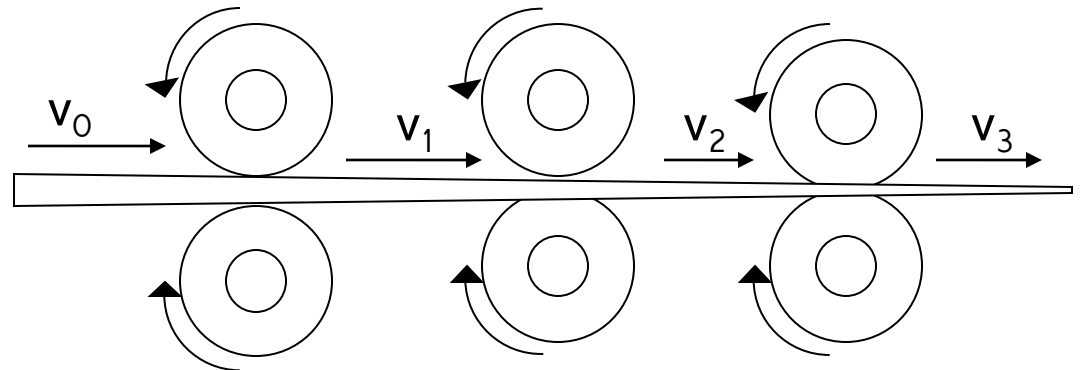
Three high



Four high



Cluster Mill



Tandem rolling mill

# Cluster Mill

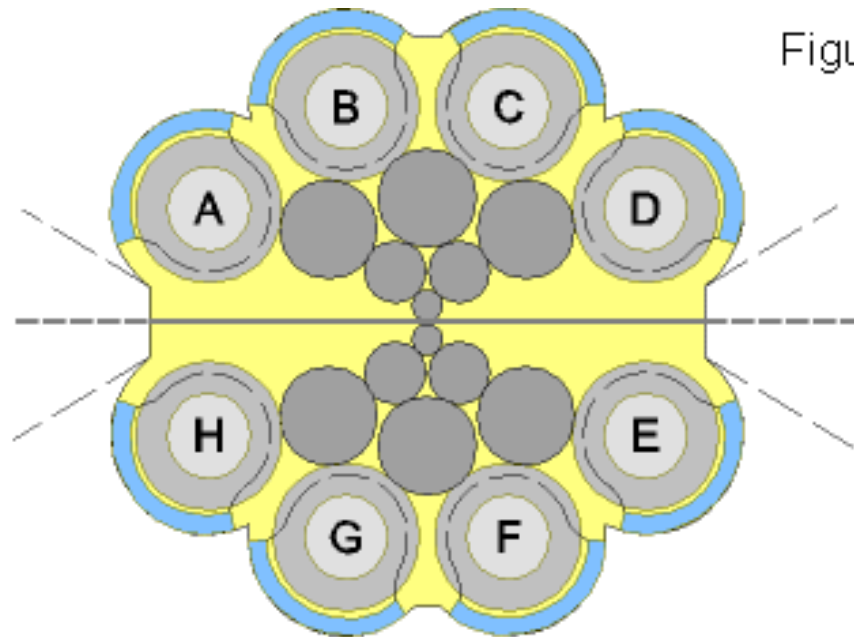


Figure 1

# Shapes Produced by rolling



Square



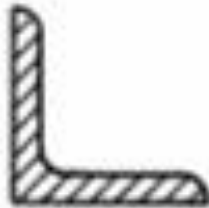
Slab



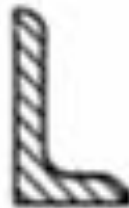
Hexagonal



Round



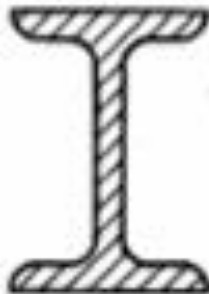
Equal-sided angle



L-section



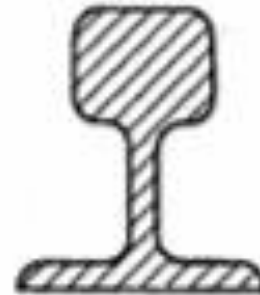
T-section



I beam



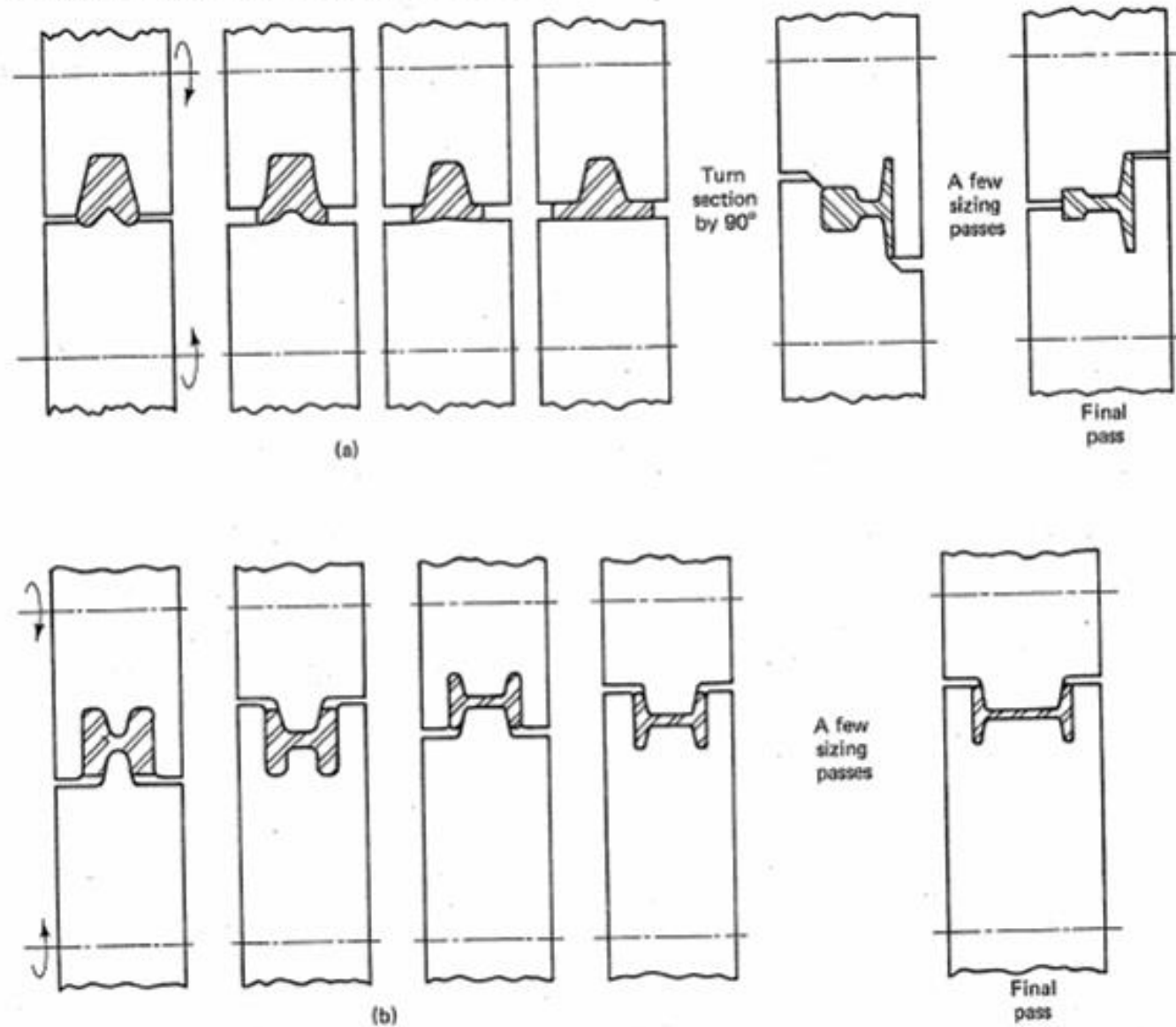
Channel beam



Rail

**FIGURE 5.10**

Roll passes: (a) for producing rails; (b) for producing an I beam



# Roll-Forging

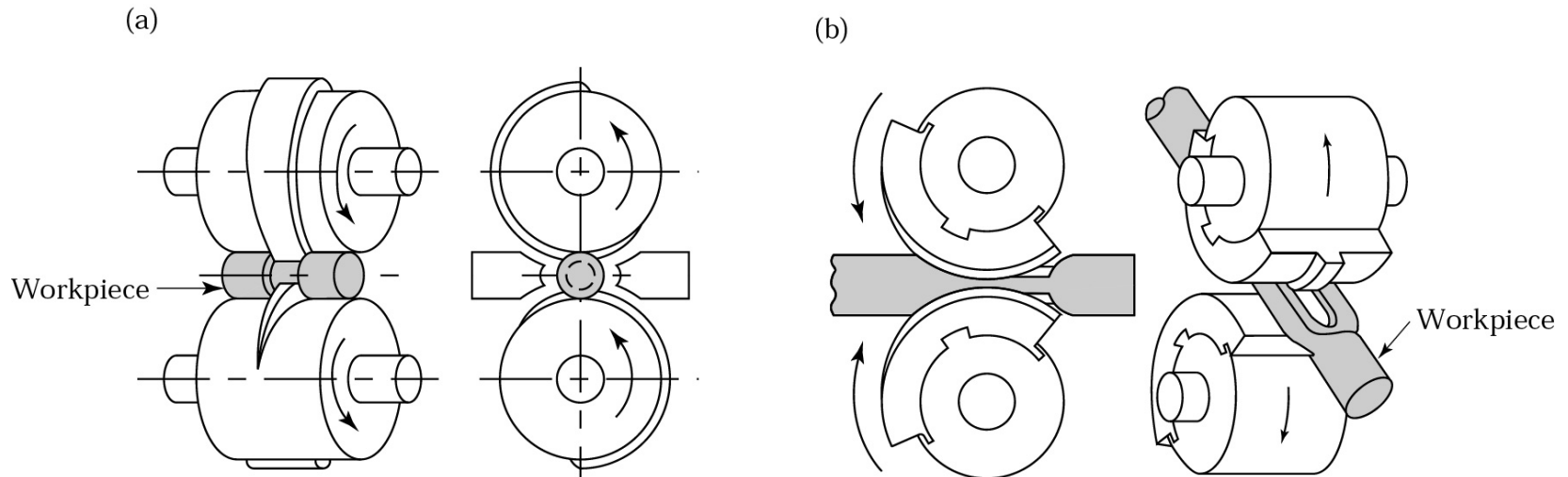
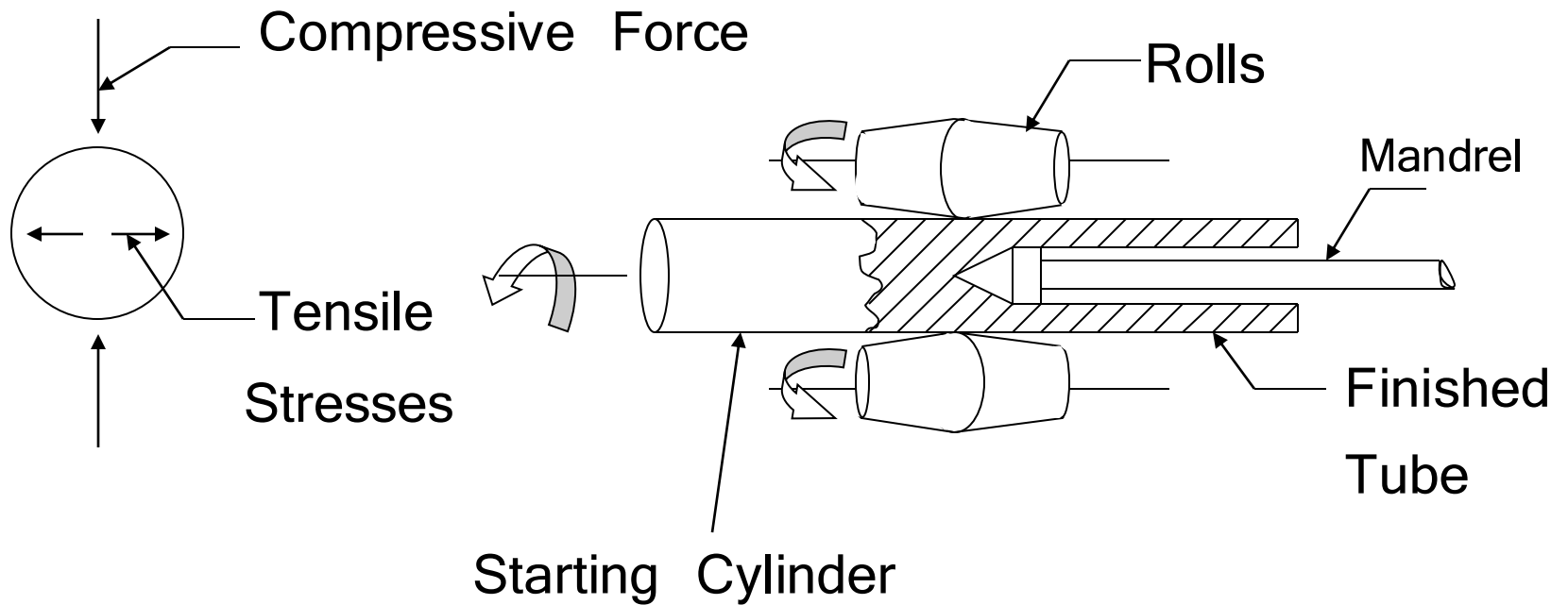


Figure 14.13 Two examples of the roll-forging operation, also known as *cross-rolling*. Tapered leaf springs and knives can be made by this process. *Source:* (a) J. Holub; (b) reprinted with permission of General Motors Corporation.

# Roll piercing



# Friction and lubrication

- Friction is undesirable;
  - Metal flow in the work is retarded
  - The forces to perform are increased
  - Rapid wear of tooling

sticking

# Lubrication

- Mineral oils
- Graphite
- Glass
- Graphite in water or mineral oil is a common for hot forging

# Sheet Metal Processes

Raw material: sheets of metal, rectangular, large

Raw material Processing: Rolling (anisotropic properties)

Processes:

Shearing

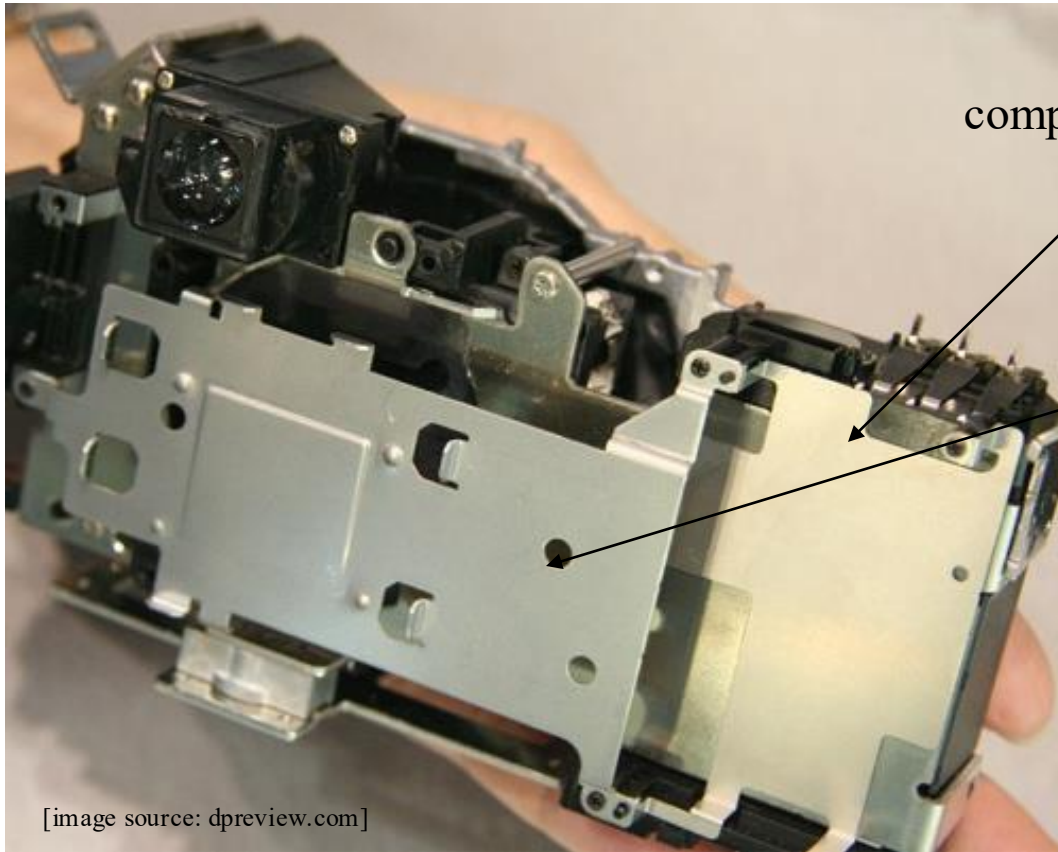
Punching

Bending

Deep drawing

# Bending

Body of Olympus E-300 camera



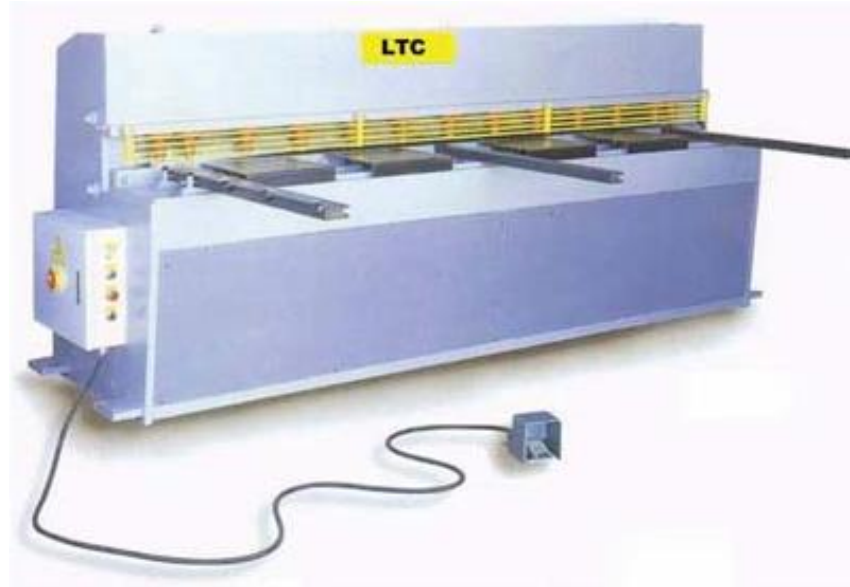
component with multiple bending operations

component with punching,  
bending, drawing operations

[image source: dpreview.com]

# Shearing

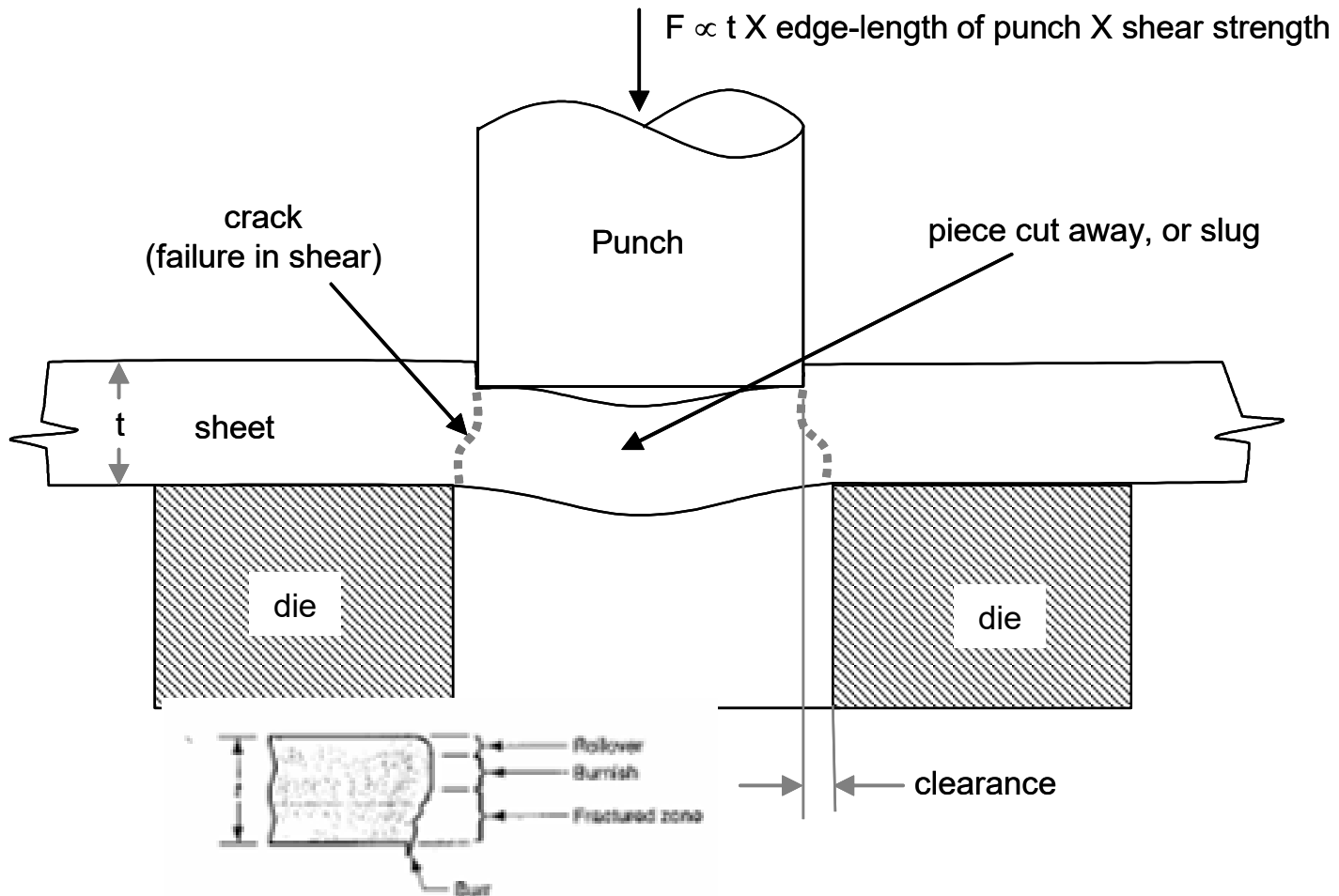
A large scissors action, cutting the sheet along a straight line



Main use: to cut large sheet into smaller sizes for making parts.

# Punching

Cutting tool is a round/rectangular punch,  
that goes through a hole, or die of same shape

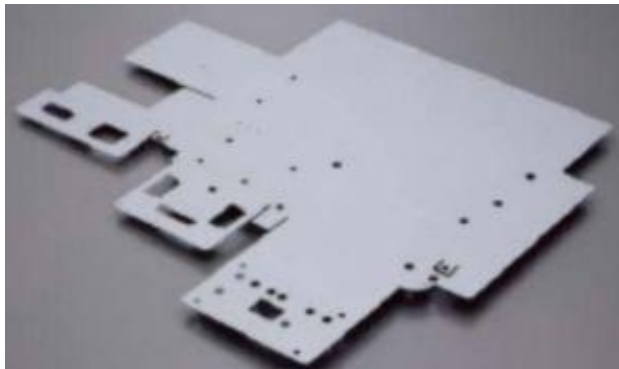


# Punching

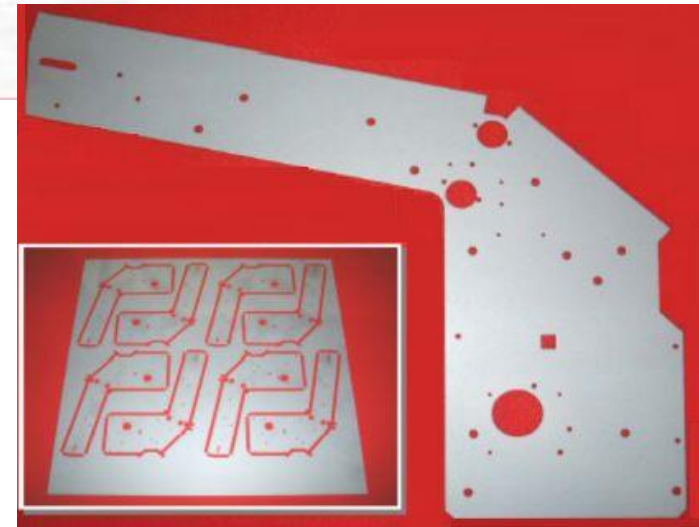
Main uses: cutting holes in sheets; cutting sheet to required shape



nesting of parts

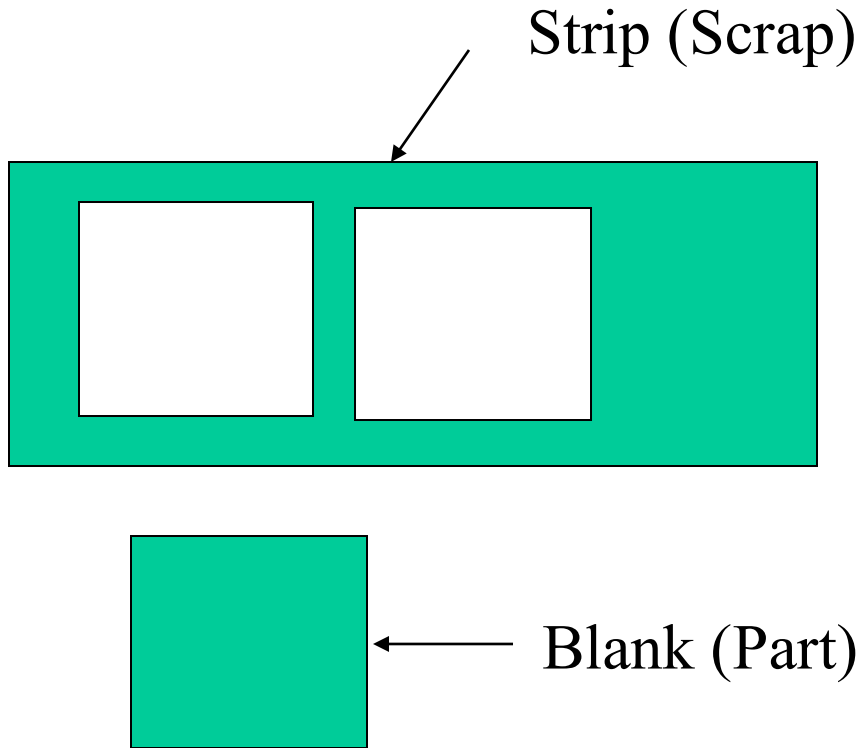


typical punched part

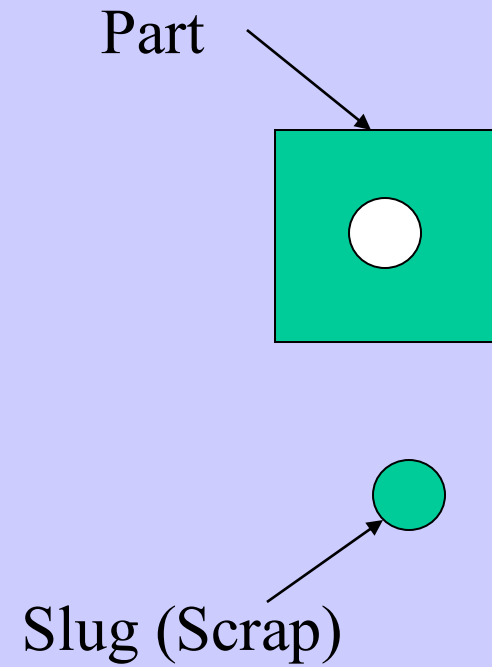


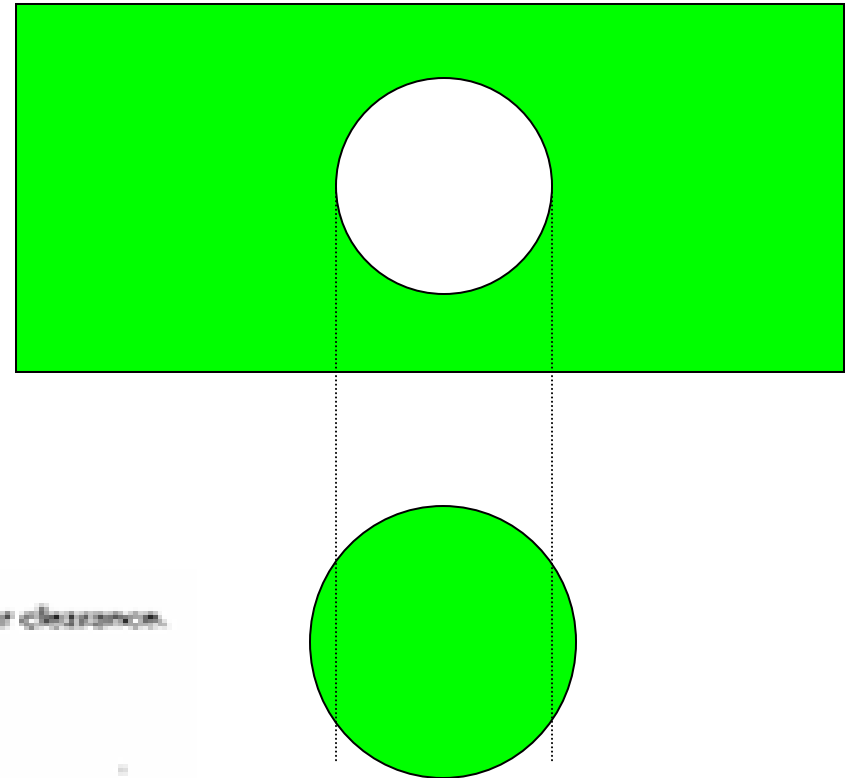
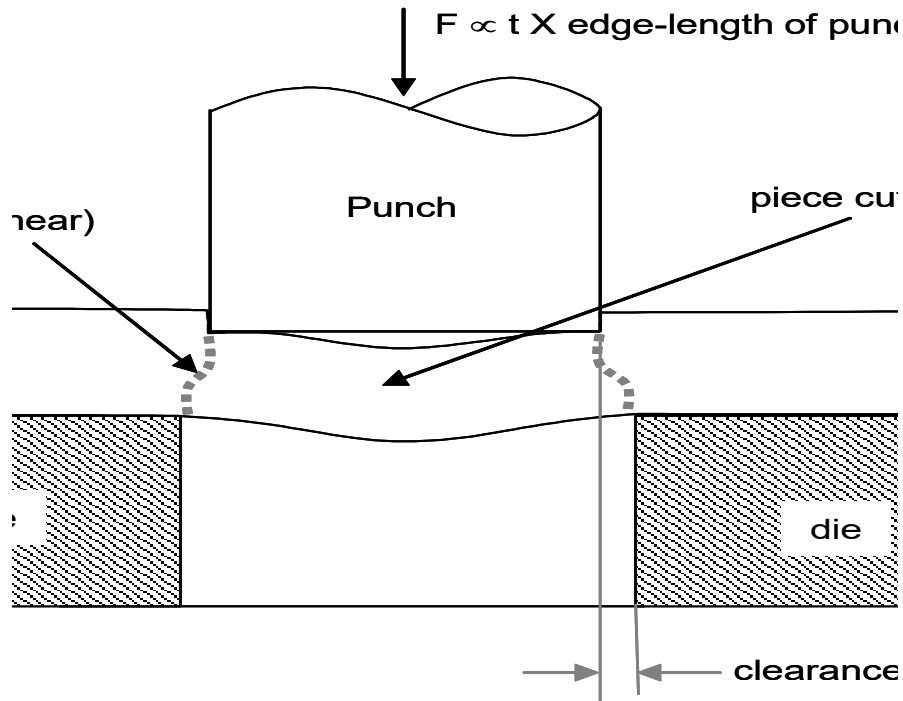
**Exercise: how to determine optimal nesting?**

# Blanking



# Piecing



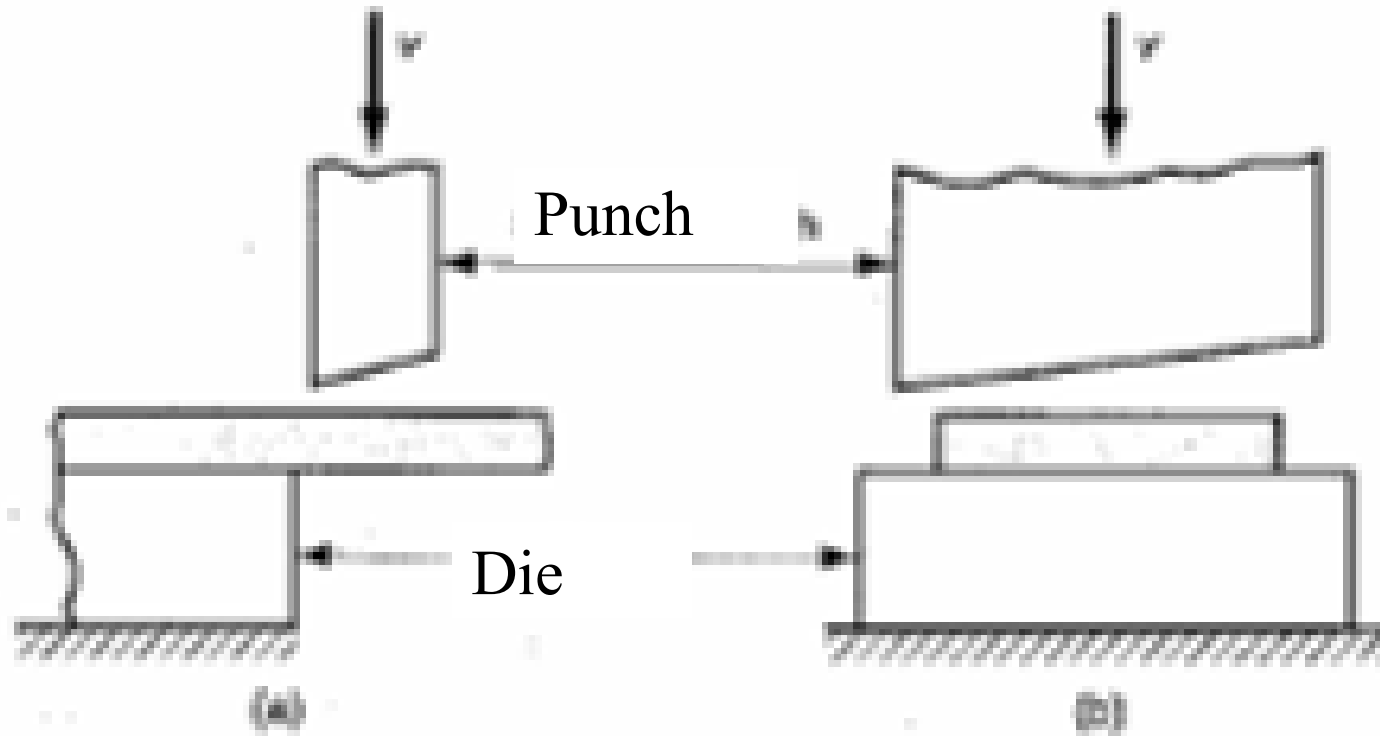


Straight portion (for resharpening)

FIGURE 22.7 Angular clearance.

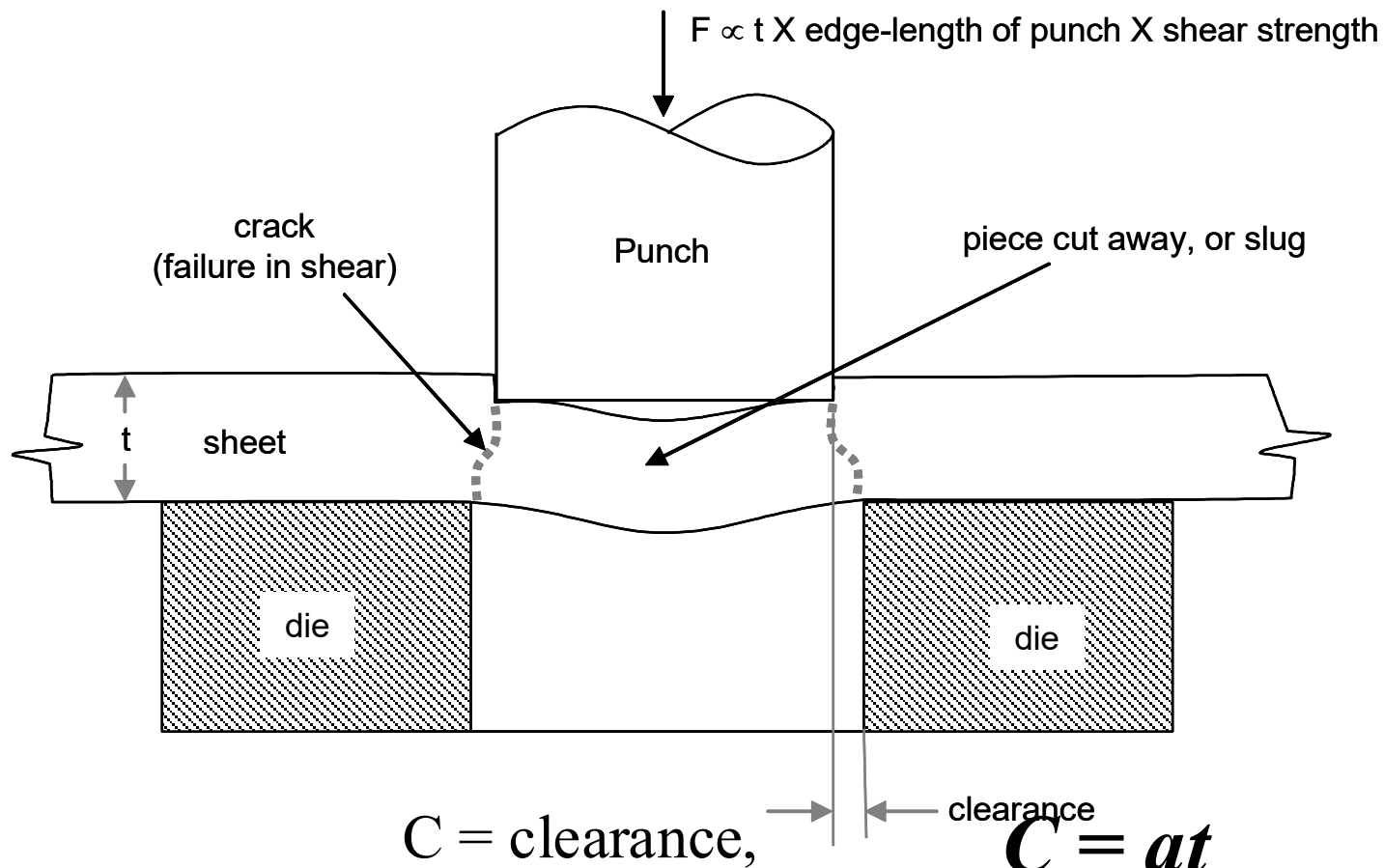


# Shearing Operation



Side View

Front View

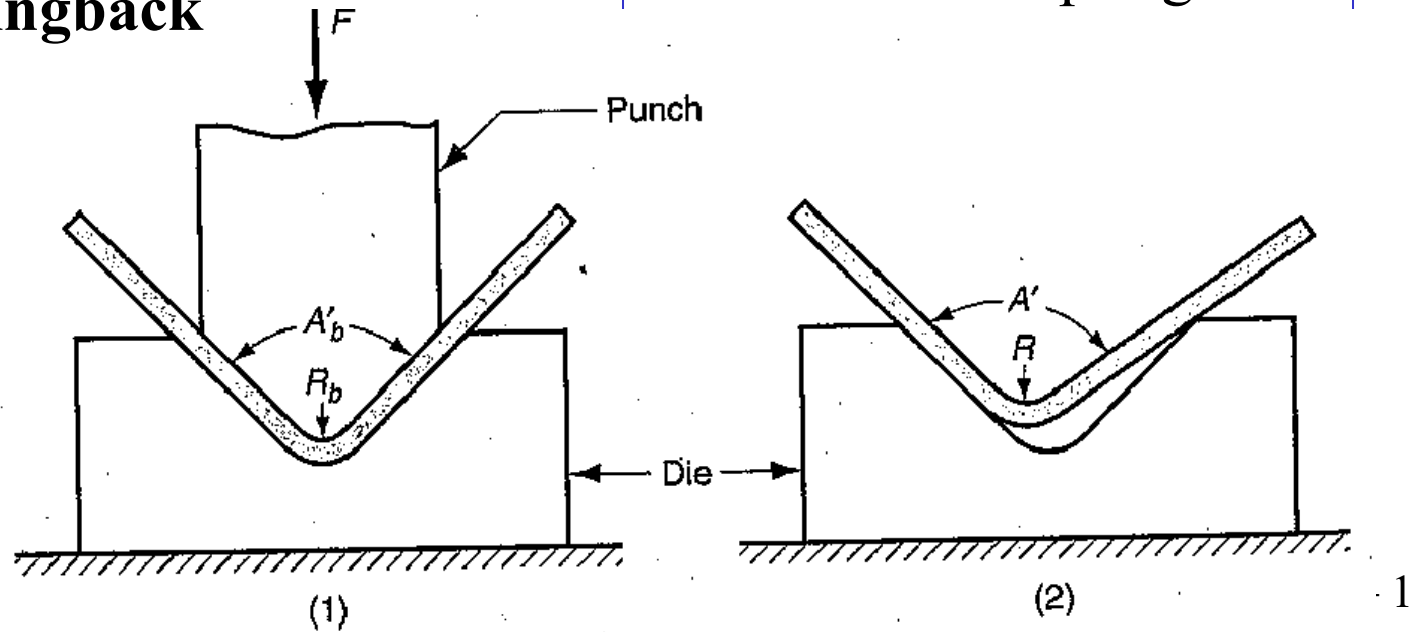


$a = \text{allowance}$

$t = \text{Thickness (plate)}$

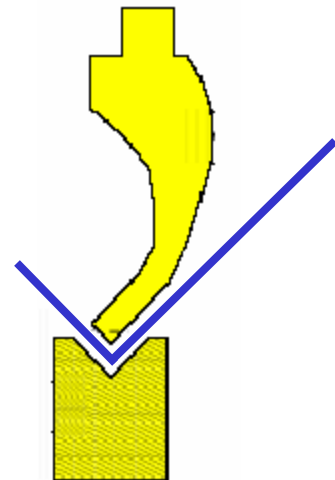
## Bending: springback

How to handle springback:



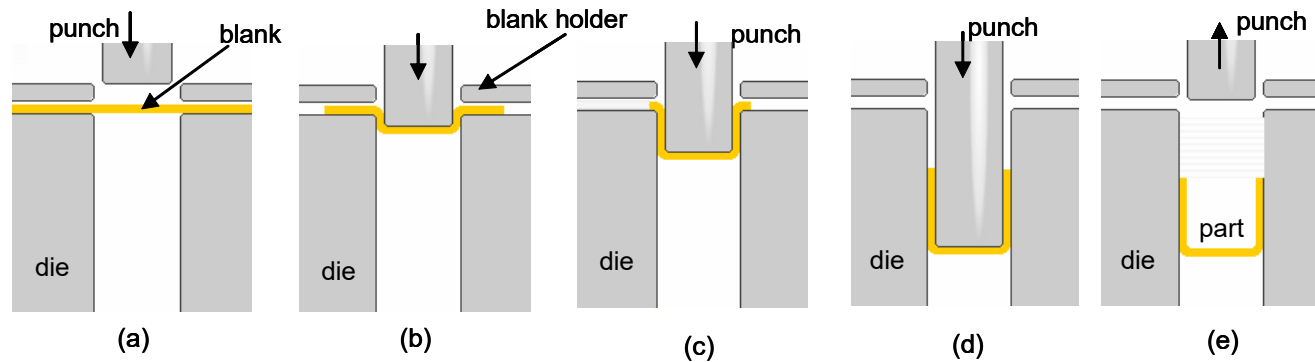
- (a) Compensation: the metal is bent by a larger angle
- (b) Coining the bend:  
at end of bend cycle, tool exerts large force, dwells

coining: press down hard, wait, release



# Deep Drawing

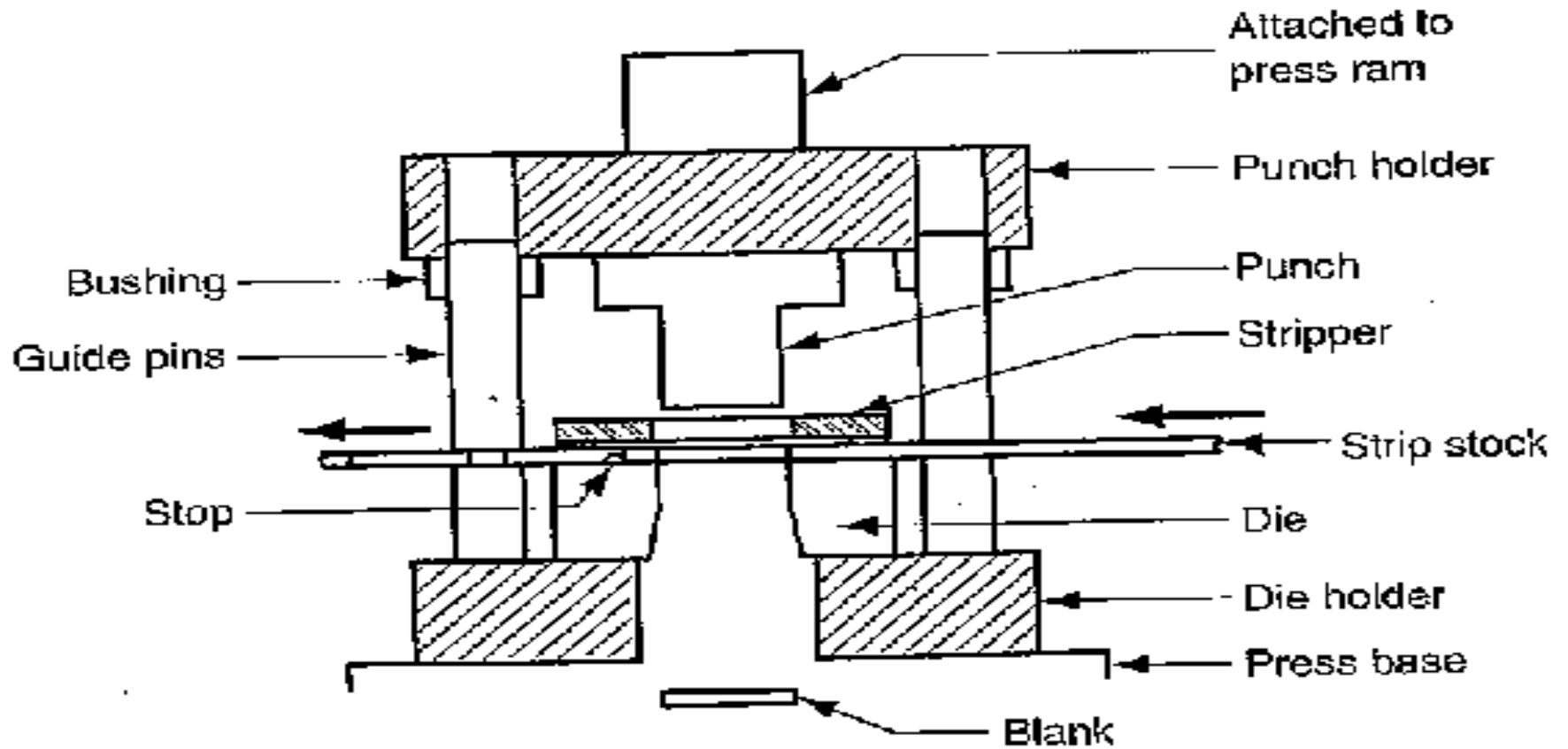
Tooling: similar to punching operation,  
Mechanics: similar to bending operation



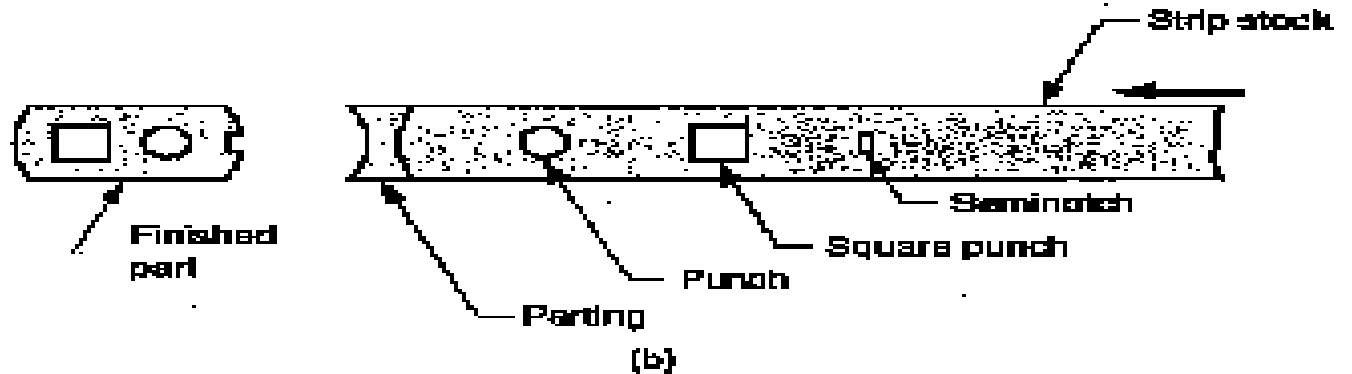
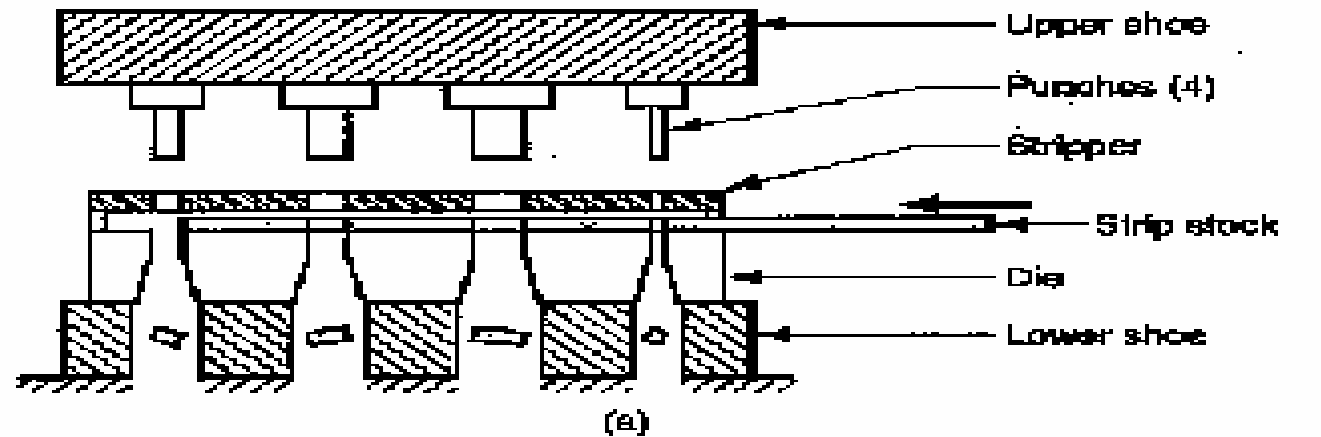
Examples of deep drawn parts

Common applications: cooking pots, containers, ...

# Punch Die



# Progressive Die



<https://youtu.be/NuFpzJLMnFs?t=88>

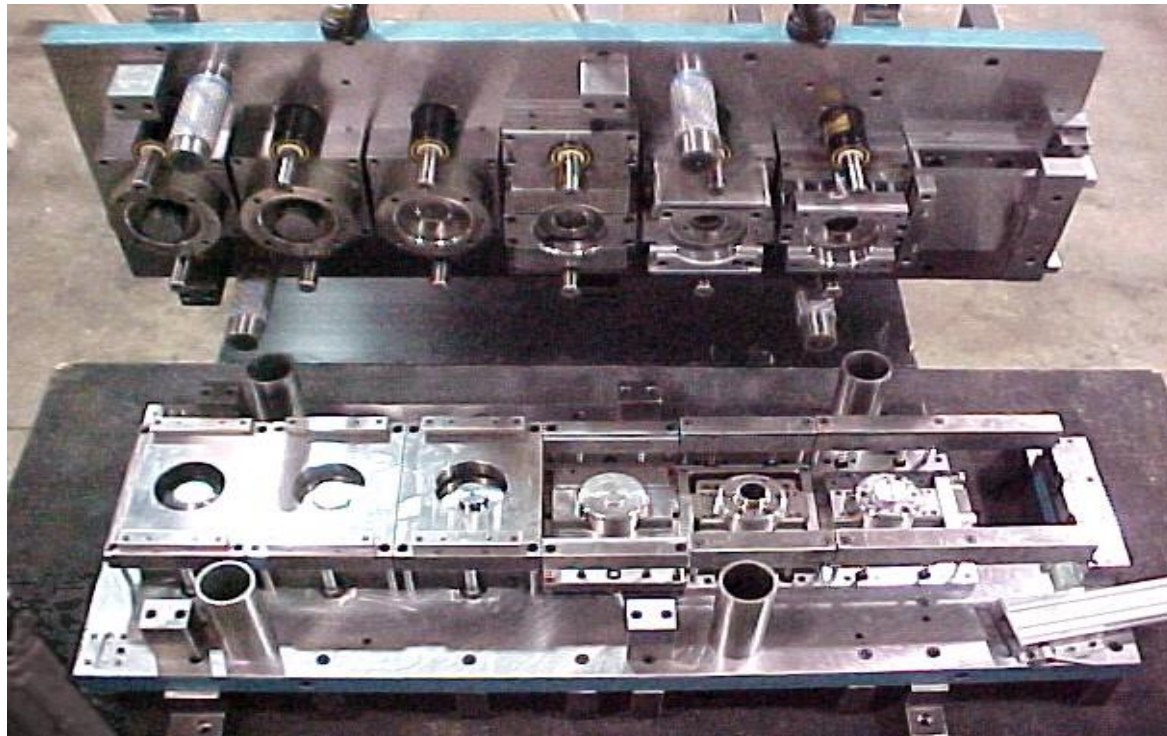
# Progressive Die Stamping

Fabricating complex parts in earlier days was an inefficient and costly affair. Either a manufacturer could repeatedly change the tool in a single press or use multiple presses each performing single action to produce one part. This resulted in long load times and a high labor cost.

With progressive die stamping, a manufacturer can eliminate all these hassles and produce better quality parts faster and in a cost-effective manner.



# Progressive Die



<https://youtu.be/10vNgC4LpkQ?t=113>

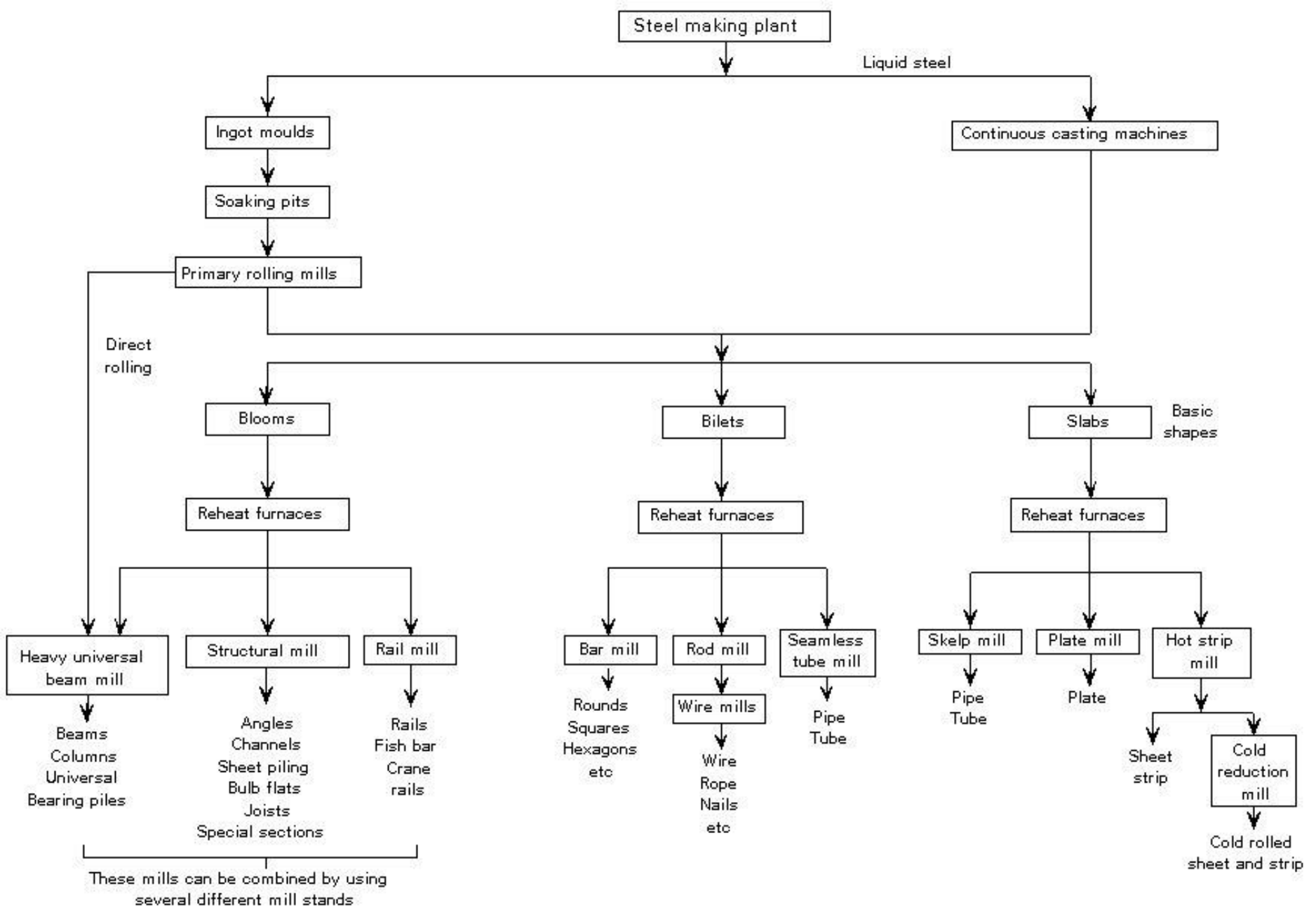


Figure 9 Principal product routes